

The Evidence Within: Evidence Sufficiency and Uptake in the Residual Stream of Retrieval-Augmented LLMs

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Abstract

Retrieval-augmented language models receive evidence piece by piece, and at some point that evidence becomes enough to answer the question. We ask whether the model’s internal state registers this moment, and whether it also registers whether the model will actually use the evidence. We build evidence trajectories for multi-hop questions: each gold paragraph is added in turn, and every decisive addition is paired with matched additions of irrelevant, duplicate or falsified paragraphs to the same starting context. Linear probes on the residual stream at the answer position, read before any token is generated, recognise the decisive addition at 0.95 AUROC in three model families and predict whether the model will take up the decisive evidence rather than ignore it at 0.78. We compare these readouts with the baselines a practitioner would otherwise use. For sufficiency, the internal state is far more reliable than the model’s own verbalised judgement (0.98 against 0.71–0.77 AUROC with paragraph count matched): the models declare complete evidence insufficient a third of the time, and the probe recovers most of those cases. For uptake, text classifiers stay near chance, whether the model answers at all already predicts much of it, and the internal state adds to that signal and is the best single predictor of correctness among the answers the model gives. The probe directions turn out to be readouts rather than levers: steering along them changes nothing, whereas the difference between the mean sufficient and insufficient states acts as a gate on abstention. The readouts replicate on HotpotQA, on RAG-Bench’s naturally retrieved documents across five domains, on a synthetic benchmark and on BM25- and E5-retrieved Wikipedia evidence, with directions learned on Wikipedia transferring only partially to other domains.

1 Introduction

Retrieval-Augmented Generation (RAG) grounds the answers of a Large Language Model (LLM) in retrieved evidence (Lewis et al., 2020). In practice the evidence rarely arrives all at once. A retriever returns passages of mixed relevance, an agent fetches them over several rounds, and for multi-hop questions the answer depends on several separate facts that may be retrieved at different times (Trivedi et al., 2022). Two questions then matter at every step. Is the evidence now *sufficient* to answer? And, if it is, will the model actually *take it up*, or will it ignore the decisive passage and answer as before? Today both questions are usually put to the model itself: it is told to say “INSUFFICIENT” when it cannot answer, and its output is taken as the verdict.

This paper asks whether the model’s internal state gives a better answer. Prior work has shown that residual-stream activations carry information about the conflict between retrieved context and parametric memory, and about how much the model relies on the context (Zhao et al., 2025, 2024). Nothing comparable exists for sufficiency and uptake, and we know of no study that compares an internal readout of these properties with the model’s own verbalised judgement, or with a text classifier given the same input.

We study this with *evidence trajectories*. For every multi-hop question we add the gold paragraphs one at a time and run a fresh forward pass on each context, reading the residual stream at the position that produces the first answer token, before anything is generated. The addition that makes the evidence complete, the *decisive increment*, is paired with controls that add, to the same starting context, a distractor of the same length, a duplicate of a paragraph already present, a falsified version of the missing paragraph, or an irrelevant paragraph that mentions the answer string. A probe trained

to tell the decisive increment from these controls cannot rely on length, on the presence of the answer string or on lexical relevance alone. Every increment is also labelled with what the model’s answer did, so a second probe, trained on decisive increments only, predicts uptake: whether the answer becomes correct, or the new evidence is ignored.

Our findings are, in order of importance, the following. First, sufficiency and uptake are both linearly readable before generation, at 0.95 and 0.78 AUROC, on splits that keep supporting documents, answer entities and relation types disjoint between training and test. Second, these readouts compare favourably with the alternatives. For sufficiency the internal state is far more accurate than the model’s verbalised judgement, which declares complete evidence insufficient a third of the time, and somewhat more accurate than a text classifier that sees the whole context. For uptake no text classifier gets far above chance; the model’s decision to answer at all predicts it about as well as the probe does, and the probe adds to that decision and is the best single predictor of whether an answer the model gives is correct. Third, as a corollary, the probe directions are not levers: steering, patching and projecting along them leave behaviour unchanged. The one direction with a causal effect is the difference between the mean sufficient and insufficient states, and it acts as a gate on abstention rather than on evidence use, which connects our results to work on steering abstention through confidence directions (Turner et al., 2024; Kumaran et al., 2026). Fourth, the readouts are recoverable in three model families and on other datasets, and directions learned on MuSiQue transfer to HotpotQA almost intact but only partially to RAGBench’s naturally retrieved documents and to synthetic evidence.

Our contributions are as follows.

- **An increment-based protocol** for reading evidence sufficiency and evidence uptake from a single forward pass before generation, with matched controls that rule out the surface explanations.
- **A comparison with the baselines that matter:** text classifiers on the same inputs, and the model’s own verbalised judgement and confidence. It shows where the internal state adds information and where it does not. As a corollary, we show that the probe directions

are readouts rather than mechanisms, and that the mean-difference direction is a steerable abstention gate.

- **Generalisation** to Llama and Mistral, to HotpotQA (Yang et al., 2018), to RAGBench (Friel et al., 2024) across five domains, to a synthetic benchmark with paraphrased, partial and contradictory evidence, and to retrieved rather than curated evidence. We release the Evidence State Transition Benchmark (ESTB), which collects these sources under one labelling scheme, together with our code.

2 Related work

3 Experimental setup

Data. We need questions whose answer depends on several separate facts, with annotations saying which paragraphs contain them. A context is then *sufficient* exactly when it contains every gold paragraph, so sufficiency is defined by the construction and not by the model’s behaviour. Multi-hop question answering provides this. Our main experiments use the 2-hop questions of MuSiQue (Trivedi et al., 2022), which composes single-hop questions so that every hop is needed, and gives one gold Wikipedia paragraph per hop, the sub-question and sub-answer of each hop, and 18 distractor paragraphs. For each model we screen 4,000 questions without any evidence and keep 2,000 that the model answers incorrectly, so that a correct answer after evidence is added can be attributed to the evidence. Section 7 adds 2WikiMultiHopQA (Ho et al., 2020), 3–4-hop MuSiQue, HotpotQA (Yang et al., 2018), RAGBench (Friel et al., 2024), whose documents were retrieved by real RAG pipelines in five domains, and a synthetic benchmark over fictional entities.

Models. Qwen2.5-7B-Instruct (28 layers, $d = 3584$), Llama-3.1-8B-Instruct (32 layers, $d = 4096$) and Mistral-7B-Instruct-v0.3 (32 layers, $d = 4096$), in bfloat16, with greedy decoding of at most 16 tokens and each model’s chat template.

Prompt and anchor. Every context is shown with the same instruction: “Use only the supplied evidence to answer the question. If the evidence is insufficient, output INSUFFICIENT.” The model therefore makes an explicit sufficiency judgement in its output, which we later use as a baseline. We

read the residual stream after every decoder block at the last prompt token, the position from which the first answer token is predicted, so all readouts are taken before generation. A second instruction that lets the model use its own knowledge as well gives the same results (Appendix H).

Evidence trajectories. For each question we build a family of contexts; Appendix A lists them all. Writing $g_1, \dots, g_n$ for the gold paragraphs, we start from the empty context and from a context of n distractors matched in length to the gold paragraphs. We then add the gold paragraphs one at a time, in the hop order of the dataset and in reverse order. The context that receives the last gold paragraph is the *minimal sufficient* context. Every context that lacks exactly one gold paragraph, which we call *penultimate*, also receives four alternative additions: a distractor of the same length as the missing paragraph, a duplicate of a gold paragraph already present, a falsified copy of the missing paragraph in which the hop’s sub-answer is replaced by another question’s sub-answer of the same type, and a distractor to which the sentence “not to be confused with {answer}” is appended. Finally, the minimal sufficient context receives a further distractor, a duplicate gold paragraph, a conflicting falsified paragraph, and the full set of 20 paragraphs. New paragraphs are inserted at a random position, so the position of the decisive paragraph carries no information. For a 2-hop question this gives 17 contexts.

Increments and tasks. An *increment* is a pair of contexts $C_{k-1} \subset C_k$ that differ by one addition. The *decisive* increment adds the last missing gold paragraph. The other increments add a gold paragraph that still leaves one missing (*non-decisive*), or something *irrelevant* to sufficiency: a distractor, a duplicate, a falsified paragraph, an answer-string control, or any addition to a context that is already sufficient. Table 1 summarises how the two tasks use them; the full list of increment kinds is in Appendix A. *Sufficiency* asks whether an increment is the decisive one. Its positives are the decisive increments and its negatives all the others; the *matched* version keeps only the four controls built from the same penultimate context, which is the harder comparison. *Uptake* is defined on decisive increments only, by what the model’s answer does: it is positive when the answer becomes correct and negative when the model still answers wrongly or says INSUFFICIENT. Correctness is exact match

Task	Positive increments	Negative increments
Sufficiency	decisive	non-decisive gold paragraph; irrelevant additions (distractor, duplicate, falsified, answer string); additions after sufficiency
matched	decisive	the four irrelevant additions made to the same penultimate context
Uptake	decisive, answer becomes correct	decisive, answer stays wrong or INSUFFICIENT

Table 1: The two tasks and the increments that define them. Table 6 in Appendix A gives the complete list.

on any alias, or containment within a short prediction. Every increment also carries the model’s full behavioural transition, which defines three further tasks, correction, stability and revision, reported in Appendix D.

Behaviour. The models follow the instruction when the evidence is clearly lacking: they say INSUFFICIENT on 94–97% of distractor-only contexts. On the minimal sufficient context they answer correctly only 36–47% of the time and say INSUFFICIENT 30–36% of the time, so the verbalised judgement is often wrong on complete evidence; a falsified final paragraph halves abstention (Table 9, Appendix F).

Splits. Probes are trained on 70% of the questions and tested on the rest. By default we use the *document split*, in which questions that share a supporting Wikipedia page are kept on the same side, so a probe cannot succeed by memorising facts about specific pages. Splits by question, by answer entity and by relation type give the same results (Appendix G).

Baselines. Two kinds of baseline are run on exactly the same increments, labels and splits. *Text classifiers* see the input but not the model: a TF-IDF logistic regression and a fine-tuned DeBERTa-v3-base cross-encoder (He et al., 2023), each given the question and the added paragraph, the latter also with the previous context (Appendix B). *Self-verbalisation* uses the model’s own output: whether it answers rather than saying INSUFFICIENT, and its confidence, measured as the top-token probability or the negative entropy at the anchor.

4 Method

Paired updates and probes. Let $h^{(l)}(C) \in \mathbb{R}^d$ be the residual stream after block l at the anchor for context C . For the increment $C_{k-1} \rightarrow C_k$ we form the paired update $\Delta h^{(l)} = h^{(l)}(C_k) - h^{(l)}(C_{k-1})$, which isolates what the added paragraph did to the model’s state and removes the content that both contexts share. For each task and layer we fit an ℓ_2 -regularised logistic regression on standardised updates with class balancing. We probe every layer and report the one selected by grouped three-fold cross-validation on the training questions only; the depth profile is in Appendix C. The probe direction is the weight vector mapped back to activation space and normalised. We also compute the *difference-of-means* direction $v_{\text{dm}} \propto \mathbb{E}[\Delta h \mid y=1] - \mathbb{E}[\Delta h \mid y=0]$ and, for comparison, probes on the raw states $h(C_{k-1})$ and $h(C_k)$ (Appendix E).

Nuisance analysis. A paired update also reflects surface properties of the increment: how much text was added, whether it mentions the answer, how lexically relevant it is to the question, where it was inserted, and how much the model’s output distribution moved. We collect these into a *nuisance vector* z per increment (the full list is in Appendix B) and run two tests. A classifier that sees only z measures how far surface properties alone go. The *gain* is the AUROC added when the probe score, computed out of fold on training increments, is given to that classifier as an extra feature; it measures what the activations contribute beyond those properties.

Generalisation. Directions are learned on the training questions of 2-hop MuSiQue and then applied, without retraining and at the layer chosen there, to other settings: a conversational protocol in which the model sees its previous answer before the new paragraph, 2WikiMultiHopQA, 3–4-hop MuSiQue, and the external benchmarks of Section 7. Where the target has its own trajectories we also train probes on it, which gives the ceiling for transfer. Hidden sizes differ across model families, so there we replicate the whole procedure instead of transferring vectors.

Causal tests. A direction from which a property can be read is not necessarily one the model uses. We therefore ask three questions of a direction v , on 150 held-out questions, each with an *insufficient* prompt that lacks one gold paragraph and the

sufficient prompt in which it has been added. *Does pushing along v change behaviour?* Steering adds $\alpha u v$ to the residual at the anchor and at every generated position, where u is the gap between the sufficient and insufficient means along v , so that $\alpha = 1$ moves the activation by one such gap; a random direction of the same norm is the control. *Does the model need v ?* Projection-out takes the sufficient prompt and sets its component along v to the insufficient-state mean, from the chosen layer onward; if the model relied on v , it should start abstaining. *Where in the update does the causal content lie?* Patching runs the insufficient prompt with its anchor residual replaced by the sufficient prompt’s, either in full, or only in the component along v , or only in the component orthogonal to v ; further variants restrict the replacement to a random subspace, to the top principal components of training updates, or to the unembedding direction of the gold answer’s first token (Appendix B). Outcomes are the rate at which the model answers rather than saying INSUFFICIENT and the rate of correct answers. We test the logistic probe direction v_{lr} and the difference-of-means direction v_{dm} (Marks and Tegmark, 2023; Turner et al., 2024; Meng et al., 2022).

5 Reading sufficiency and uptake before generation

Every number in this section is the AUROC of a probe on held-out questions of the document split, at the layer chosen by cross-validation on the training questions.

Sufficiency. A probe trained to recognise the decisive increment separates it from all other increments at 0.948–0.952 AUROC in the three models (Table 2). Against the matched controls alone, which start from the same penultimate context and add a distractor, a duplicate, a falsified paragraph or an answer-string control instead of the missing paragraph, it still reaches 0.91–0.92. Surface properties do not explain this. A classifier on the nuisance vector reaches 0.86–0.87, the probe adds 0.11–0.12 on top of it, and a probe trained after the nuisance covariates have been regressed out of the activations still reaches 0.94–0.96. The result is the same under question-, answer- and relation-disjoint splits (0.94–0.96, Appendix G). The signal is strongest at 50–60% of the network’s depth, and projections onto the direction order the increment kinds as one would expect if it encoded

Task	Qwen2.5-7B			Llama-3.1-8B			Mistral-7B		
	AUROC [95% CI]	nuis.	gain	AUROC [95% CI]	nuis.	gain	AUROC [95% CI]	nuis.	gain
Sufficiency	0.948 [0.94,0.95]	0.87	0.11	0.952 [0.95,0.96]	0.86	0.12	0.949 [0.94,0.95]	0.87	0.11
Sufficiency (matched)	0.911 [0.90,0.92]	0.90	0.07	0.918 [0.91,0.93]	0.89	0.08	0.910 [0.90,0.92]	0.90	0.07
Uptake	0.775 [0.74,0.80]	0.66	0.14	0.783 [0.75,0.81]	0.58	0.21	0.771 [0.74,0.80]	0.59	0.18

Table 2: Held-out AUROC of the probe on paired updates Δh (document split; 95% question-level bootstrap intervals). “nuis.”: a classifier that sees only surface properties of the increment (length change, presence of the answer string, lexical relevance, change in output confidence). “gain”: AUROC added when the probe score is given to that classifier.

“the evidence has just become complete”: decisive first, then falsified, distractor, duplicate and answer-string additions, with additions to an already sufficient context far below (Appendix C).

Uptake. Among decisive increments, a second probe predicts whether the model will use the new paragraph, that is, whether its answer becomes correct, at 0.77–0.78 (0.85–0.87 on the question split), at layers near the end of the network. This is the task on which the activations add most beyond surface properties: the nuisance classifier reaches only 0.58–0.66 and the probe adds 0.14–0.21. Uptake is a different signal from sufficiency. The two directions are nearly orthogonal (cosine 0.02–0.03), and the sufficiency direction, used to predict uptake, scores only 0.54–0.60.

State or update? One may ask whether the subtraction in Δh matters. Refitting the probes on the raw state after the addition, $h(C_k)$, gives 0.93–0.94 for sufficiency and 0.77–0.80 for uptake, and a probe on the concatenation of both states matches Δh ; the state before the addition is uninformative beyond the stage of the trajectory it belongs to (Appendix E). The information therefore sits in the model’s state after it has seen the evidence. The paired update is a controlled way of reading it: it discards what both contexts share and lets every decisive increment be compared with controls from the same starting point.

5.1 Do internal states beat the obvious baselines?

A practitioner who wants to know whether the evidence is sufficient, or whether the model will use it, has two cheaper options: a text classifier on the same input, or the model’s own output, which under our instruction is an explicit verdict on sufficiency. Table 3 compares both with the probe on the same test increments.

Text classifiers. For sufficiency, TF-IDF reaches 0.62 and a DeBERTa that sees only the question and the added paragraph 0.72–0.73. A DeBERTa that also sees the previous context reaches 0.89–0.92, below the probe’s 0.95 but not by much: sufficiency is largely a property of the input, and the model’s state is an accurate readout of it. Uptake is the opposite case. No text classifier exceeds 0.60 on the same increments (0.63 with a larger training budget), against 0.77–0.78 for the probe. Whether the model will use a paragraph is not visible in the paragraph.

The model’s own judgement. The verbalised judgement is a weak sufficiency detector. Whether the model answers after the increment separates decisive from other increments at only 0.65–0.69 (0.67–0.73 on the matched task), because the models say INSUFFICIENT on a third or more of complete contexts (Section 3). Output confidence is no better; it is in fact slightly *lower* after the decisive paragraph than after a control, since INSUFFICIENT is a confident output. The internal state is far ahead. A probe that classifies single contexts as sufficient or not, read at the same anchor of the same forward pass and restricted to two-paragraph contexts so that paragraph count cannot help, reaches 0.976–0.986 against 0.71–0.77 for the verbalised judgement, and it stays at 0.967–0.982 among the contexts the model rejects. With its threshold set for 0.95 precision on training data, it recovers 76–81% of the complete contexts the model wrongly called insufficient, with 3–4% false alarms on the incomplete ones it rightly rejected. The update probes behave the same way when the output is held fixed: trained and tested only on increments after which the model still says INSUFFICIENT, they reach 0.935–0.944, against 0.65–0.69 for the output itself. The sufficiency readout is therefore not the model’s planned output. The residual stream registers that the evidence

Model	Task	text classifiers			the model's own output		
		TF-IDF	DeBERTa (q+added)	DeBERTa (+context)	verbalised	confidence	probe
Qwen2.5-7B	sufficiency	0.623	0.723	0.900	0.685	0.428	0.948
	sufficiency (matched)	0.623	0.758	0.823	0.721	0.408	0.911
	uptake	0.574	0.569	0.557	0.776	0.571	0.775
Llama-3.1-8B	sufficiency	0.625	0.715	0.888	0.692	0.409	0.952
	sufficiency (matched)	—	—	—	0.726	0.390	0.918
	uptake	0.463	0.466	0.517	0.783	0.566	0.783
Mistral-7B	sufficiency	0.625	0.730	0.920	0.646	0.452	0.949
	sufficiency (matched)	—	—	—	0.671	0.428	0.910
	uptake	0.536	0.573	0.604	0.775	0.609	0.771

Table 3: Baselines on the same test increments as the probe (document split, AUROC). Text classifiers see the question and the added paragraph (“q+added”) or also the previous context (“+context”). “verbalised”: the model answers rather than saying INSUFFICIENT after the increment. “confidence”: the better of top-token probability and negative entropy at the anchor. The matched task has text baselines for Qwen only.

is complete even when the answer denies it (Appendix F).

For uptake the picture is more nuanced, and Table 3 shows it plainly. Whether the model answers at all predicts uptake at 0.78, the same AUROC as the probe, because about half of the decisive increments the model fails to take up are ones after which it says INSUFFICIENT. The informative comparison is among the decisive increments the model does answer, where the question becomes whether the answer is right. There, the probe separates correct from wrong answers at 0.63–0.68 and output confidence at 0.63–0.64, with the two nearly identical on Mistral (Table 12 in Appendix F). Combining the decision to answer with the probe gives 0.84–0.86, against 0.84 with confidence. So the internal state does carry uptake information that neither the text nor the model’s output has, but much of what the uptake probe reads at 0.78 is the model’s own decision to answer, and the part specific to correctness is modest.

6 Readouts, not levers: causal tests

The probes of Section 5 read sufficiency and uptake accurately. We now ask, as a corollary, whether the directions they read are also the ones the model uses to decide. The tests of Section 4 are applied to the logistic probe direction v_{lr} and to the difference-of-means direction v_{dm} ; Table 4 gives the results.

Probe directions are inert. Steering along the logistic sufficiency or uptake direction changes no rate at any strength from 0.7 to 22 units of ℓ_2 norm, up to 35% of the typical norm of the resid-

ual stream, and a random direction of the same norm is equally inert, so the null result is not a matter of dose (Appendix N). Patching only the component of the sufficient-minus-insufficient update that lies along the probe direction leaves the insufficient prompt’s behaviour unchanged in all three models (“ $\parallel v$ ” in Table 4), and projecting the direction out of sufficient prompts changes nothing either (v_{lr} column).

The anchor residual is causally sufficient, but outside the probe direction. Replacing the insufficient prompt’s anchor residual with the sufficient prompt’s takes Qwen from 0.04/0.13 (correct/answered) to 0.26/0.71 and Llama from 0.02/0.11 to 0.24/0.74, about half the correctness the model reaches when the paragraph is actually present; Mistral shows the same pattern with a weaker effect (0.09/0.60). Removing the probe component from the patch leaves the effect intact (“ $\perp v$ ”). The single anchor position carries enough to make the model answer, often correctly, but none of that content lies along the direction the probe reads. The content is also high-dimensional: restricted to the top principal components of training updates, the effect grows gradually with their number, and a random 256-dimensional subspace does nothing (Appendix P).

The mean-difference direction is an abstention gate. The one direction with a causal effect is v_{dm} , which the logistic probe does not align with. Setting its component to the insufficient-state value makes all three models abstain on essentially every sufficient prompt (answered 0.01, 0.00, 0.01, against 0.72, 0.73, 0.63 for a random direction treated identically), and equally on

Model	Patching the insufficient prompt (correct/answered)								Projection-out on sufficient (answered)		
	none	full	$\parallel v$	$\perp v$	rand-256	PC1	PC256	unemb.	v_{dm}	random	v_{lr}
Qwen2.5-7B	0.04/0.13	0.26/0.71	0.04/0.13	0.25/0.71	0.04/0.13	0.09/0.29	0.18/0.59	0.09/0.19	0.01	0.72	0.72
Llama-3.1-8B	0.02/0.11	0.24/0.74	0.02/0.12	0.24/0.73	0.02/0.12	0.02/0.19	0.09/0.41	0.19/0.62	0.00	0.73	0.75
Mistral-7B	0.07/0.17	0.09/0.60	0.07/0.17	0.09/0.60	0.07/0.17	0.08/0.61	0.08/0.56	0.07/0.19	0.01	0.63	0.63
Qwen 3-4 hop	0.05/0.20	0.25/0.62	0.06/0.19	0.25/0.62	0.06/0.20	0.07/0.23	0.20/0.51	0.07/0.21	0.03	0.63	0.63

Table 4: Causal tests on 150 held-out questions, intervening at every layer from the best sufficiency layer onward. Left: the insufficient prompt with its anchor residual patched from the sufficient prompt (correct / answered rate). “none”: no patch; “full”: the whole residual; $\parallel v$ and $\perp v$: only the component along, or orthogonal to, the probe direction; “rand-256”: a random 256-dimensional subspace; PC1, PC256: the top 1 or 256 principal components of training updates; “unemb.”: the gold answer’s unembedding direction. Right: the sufficient prompt with the component along a direction set to its insufficient-state value (answered rate). v_{dm} : difference-of-means direction; v_{lr} : logistic-probe direction.

prompts with falsified evidence. Steering along it at low strength gives a graded, monotone response in both directions, while the random control stays flat (Fig. 1; full numbers in Appendix O). At $\alpha = -1$ Qwen answers 34% of sufficient prompts instead of 71%, with unchanged precision among the answers it still gives. The positive direction, however, is indiscriminate. At $\alpha = +1$ the models answer more often on distractor-only prompts (Qwen 0.03 $\rightarrow$ 0.12, Llama 0.03 $\rightarrow$ 0.29, Mistral 0.03 $\rightarrow$ 0.28), on insufficient prompts (0.13 $\rightarrow$ 0.26, 0.11 $\rightarrow$ 0.51, 0.17 $\rightarrow$ 0.47) and on answer-string controls, and Llama follows falsified evidence twice as often (0.19 $\rightarrow$ 0.39), while correct answers on sufficient prompts gain only 3 points on Qwen and 18 on Llama. A direction that makes the model accept any passage governs whether it answers, not whether it integrates evidence. The sufficiency readout and the abstention decision are thus carried by different directions, which fits the observation of Section 5.1 that the state registers complete evidence even when the model refuses to answer.

7 Generalisation across models, datasets and evidence sources

Table 2 already shows the readouts in three model families, each with its own probes. This section asks whether the directions depend on MuSiQue’s construction, on curated Wikipedia paragraphs, or on the stateless protocol in which every context is processed afresh.

Other protocols and datasets, without retraining. Applied unchanged, the 2-hop MuSiQue directions reach 0.79 (sufficiency) and 0.78 (uptake) in a conversational protocol in which the model sees its previous answer before the new paragraph and is told to change it only if the evidence war-

rants it, so the transition is not an artefact of the stateless design. On 2WikiMultiHopQA the sufficiency direction reaches 0.96 (0.93 against matched controls) and uptake 0.79, against 0.99 and 0.91 for probes trained on 2WikiMultiHopQA itself. On 3-4-hop MuSiQue questions sufficiency transfers at 0.82-0.85; probes trained there reach 0.93 on every split, while their uptake readout is less stable across splits (0.61-0.80). Details are in Appendices I to L.

External benchmarks. We then repeat the whole pipeline, unchanged and with Qwen2.5-7B, on HotpotQA, RAGBench and the synthetic benchmark (Table 5). Sufficiency is decodable on every source (0.98, 0.90, 0.97) and against matched controls (0.96, 0.94, 0.95); on the synthetic data these controls include paraphrased duplicates, partial evidence and contradictions, and the increments are ordered along the direction as on MuSiQue. Uptake is decodable on all three (0.73, 0.69, 0.93). It is not an artefact of multi-hop questions: it reaches 0.70 on RAGBench single-hop questions, 0.79 on HotpotQA comparison questions, whose two facts are independent lookups, and 0.99 on single-hop synthetic questions, and it is above chance in every RAGBench domain (0.56-0.78; Appendix S). On HotpotQA, where we keep questions the model can answer without evidence, sufficiency is decoded equally well whether the model knows the answer closed-book or not (0.984 against 0.978), so the readout is not a signal that the model “needs” evidence it lacks.

Transfer across construction and domain. The MuSiQue directions transfer to HotpotQA at 0.95 for sufficiency, but only partially to RAGBench (0.71; matched 0.78; uptake 0.55) and to

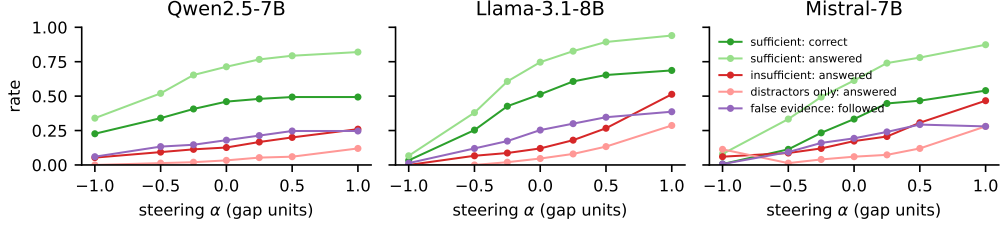


Figure 1: Dose-response of difference-of-means steering (four layers from the best sufficiency layer, from the anchor onward; α in units of the sufficient-minus-insufficient gap). Negative α induces abstention everywhere; positive α induces answering everywhere, including on distractors and false evidence.

Dataset	within-dataset probes (AUROC)				MuSiQue dir.		patched insufficient (answered)				proj.-out (answered)	
	suff.	matched	uptake	corr.	suff.	uptake	none	full	$\parallel v$	$\perp v$	v_{dm}	random
HotpotQA	0.979	0.961	0.729	0.846	0.952	0.649	0.41	0.92	0.41	0.92	0.17	0.92
RAGBench	0.898	0.944	0.685	0.742	0.706	0.550	0.15	0.72	0.15	0.72	0.10	0.73
RAGBench (held-out sub-datasets)	0.717	0.872	0.596	0.719	0.706	0.550	—	—	—	—	—	—
Synthetic	0.965	0.948	0.930	0.925	0.739	0.722	0.03	0.89	0.03	0.89	0.00	0.99
Synthetic (held-out relation chains)	0.945	0.925	0.883	0.897	0.739	0.722	—	—	—	—	—	—

Table 5: External benchmarks (Qwen2.5-7B). Within-dataset probes trained and tested on the row’s dataset; MuSiQue 2-hop directions applied without retraining; and the patching and projection-out tests of Section 6 (answered rate, 150 held-out questions). “Held-out sub-datasets” and “held-out relation chains” test on groups never seen in training.

the synthetic data (0.74; uptake 0.72). Within RAGBench, holding out whole sub-datasets at test time drops sufficiency from 0.90 to 0.72 and uptake from 0.69 to 0.60; holding out relation chains on the synthetic data costs two to five points. The state is present under natural retrieval and in every domain, but its direction is partly dataset-specific: a detector trained on Wikipedia should not be deployed on biomedical or technical RAG without adaptation. The causal picture of Section 6 replicates on all three sources: the full anchor patch makes the insufficient prompt answer (0.41 $\rightarrow$ 0.92, 0.15 $\rightarrow$ 0.72, 0.03 $\rightarrow$ 0.89), the component along the probe direction does nothing, and projecting out v_{dm} silences answering (0.17, 0.10, 0.00).

Retrieved and rewritten evidence. In a real RAG system the evidence comes from a retriever, not from a benchmark’s curated paragraphs. We therefore rebuild the trajectories of 1,200 MuSiQue questions with passages retrieved from the 21M-passage Wikipedia corpus distributed with FlashRAG (Jin et al., 2025; Karpukhin et al., 2020), by BM25 (Robertson and Zaragoza, 2009) or by the dense retriever E5 (Wang et al., 2022), either keeping the gold paragraphs and retrieving all distractors, or retrieving all the evidence, which is possible for the 402 (E5) and 219 (BM25) questions for which retrieval

found every hop. A further run varies the wording of the decisive paragraph with LLM paraphrases, value-withheld versions and paragraphs on the same entity written independently for HotpotQA or 2WikiMultiHopQA. Nothing changes qualitatively (Table 22; construction and full results in Appendix T). Sufficiency stays at 0.91–0.95, the MuSiQue directions transfer at 0.91–0.94 (sufficiency) and 0.76–0.82 (uptake) on questions the source probes never saw, and the causal dissociation reproduces in every setting. Paraphrased and independently written decisive paragraphs project like the originals; partial and contradictory versions are the hardest controls, separated from decisive increments at only 0.81 and 0.76.

8 Conclusion

We asked whether a language model’s internal state shows that newly retrieved evidence has become sufficient, and whether the model will use it. Both are readable from the residual stream at the answer position before generation, in three model families and under splits that keep documents, answers and relations disjoint. For sufficiency the internal state is a far better judge than the model’s own output, which declares complete evidence insufficient a third of the time, and somewhat better than a text classifier given the whole context. For uptake, text alone says almost nothing, the model’s

decision to answer says much, and the internal state adds to it. The directions that carry these readouts are not the ones the model acts on; the only causal direction we found is a gate on abstention. The readouts transfer across datasets, fully to HotpotQA and partially to other domains and to retrieved evidence. Beyond what they say about the model, these readouts have practical uses: read before generation, the sufficiency score can decide when an adaptive retrieval system should stop fetching evidence, and the uptake score whether to trust the answer it then gives (Appendix R). We release the trajectory construction, the probing and intervention code, and the benchmark.

Limitations

Our main experiments use questions the models cannot answer without evidence, under an instruction that asks for INSUFFICIENT when the evidence is lacking; the verbalised baseline depends on that instruction, while the internal readout changes little without it (Appendix H). The uptake probe reads much of the model’s decision to answer, and its advantage over output confidence among answered increments is small. The causal experiments on the external sources use Qwen only, RAGBench correctness relies on an LLM judge, and directions learned on Wikipedia transfer only partially to other domains. We tested additive, patching and projection interventions at the anchor position only.

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A Evidence conditions and increments

This appendix lists every context (*condition*) shown to the model for one question, and every pair of conditions (*increment*) on which probes are trained. Consider a question with n gold paragraphs $g_0, \dots, g_{n-1}$, one per reasoning hop, in the order given by the dataset’s decomposition of the question, together with its pool of distractor paragraphs. The conditions are:

- **Empty and distractor-only contexts.** C_0 contains no evidence. C_D contains n distractors, each matched in word count to one gold paragraph.
- **Chains.** Gold paragraphs are added one at a time, in the canonical hop order, in the reverse order and, for $n \geq 3$, in random orders. Each chain gives the prefixes $C_{o_1}, C_{o_1 o_2}, \dots$, and its last prefix is the minimal sufficient context. Each new paragraph is inserted at a uniformly random position among the paragraphs already present, so that the position of the decisive paragraph carries no information.

Identical contexts reached through different orders are stored once.

- **Leave-one-out contexts** ($n \geq 3$): all gold paragraphs but one, shuffled, followed by their completion with the missing paragraph.
- **Matched controls.** From every *penultimate* context S , i.e. a chain or leave-one-out context that lacks exactly one gold paragraph, we build four controls alongside the decisive completion: $S+D$ adds a distractor matched in length to the missing paragraph; $S+R$ adds a verbatim duplicate of a gold paragraph already in S ; $S+F$ adds the missing paragraph with its sub-answer (and title, if applicable) replaced by another question’s sub-answer for the same relation, so the false paragraph remains plausible in type (questions whose sub-answer does not appear in the paragraph have no such control); $S+A$ adds a distractor with the sentence “(Not to be confused with {answer}.)” appended, so the answer string is present without the supporting fact.
- **Post-sufficiency contexts.** The canonical minimal sufficient context plus a distractor, plus a duplicate gold paragraph, or plus a falsified copy of the final hop that conflicts with it; and the full context of all 20 paragraphs, shuffled.

For 2-hop questions this gives 17 conditions and 17 increments per question. Table 6 names the kinds of increment, as they appear in the tables and figures, groups them into the three families used in the main text (decisive, non-decisive, irrelevant), and shows which task uses each kind as a positive (+), as a negative (−) or not at all (·). For uptake the label is behavioural ($\pm$: positive if the answer becomes correct). The conversational protocol (Appendix J) re-runs the 10 chain-related increment kinds with the model’s own stateless answer shown in the prompt (30,000 increments for Qwen).

B Probing, nuisance and intervention details

Activation capture. Activations are read at the anchor position by forward hooks on the output of each decoder block. Prompts are processed in batches, left-padded so that the anchor has the

Family	Kind	What the increment adds	Suff.	Matched	Uptake
decisive	decisive	the last missing gold paragraph of a chain	+	+	$\pm$
decisive	loo_decisive	the missing gold paragraph of a leave-one-out context ($n \geq 3$)	+	+	$\pm$
non-decisive	gold_nondecisive	a gold paragraph that still leaves another one missing	—	.	.
irrelevant	distractor_from_none	distractors added to the empty context	—	.	.
irrelevant	distractor	a length-matched distractor added to a penultimate context ($S+D$)	—	—	.
irrelevant	redundant	a duplicate of a gold paragraph already present ($S+R$)	—	—	.
irrelevant	false	the missing paragraph with a substituted sub-answer ($S+F$)	—	—	.
irrelevant	answer_ctrl	a distractor that mentions the answer string ($S+A$)	—	—	.
irrelevant	post_distractor	a distractor added to the minimal sufficient context	—	.	.
irrelevant	post_redundant	a duplicate gold paragraph added to the minimal sufficient context	—	.	.
irrelevant	post_false	a falsified copy of the final hop, conflicting with it, added to the minimal sufficient context	—	.	.
irrelevant	full_context	all remaining paragraphs added to the minimal sufficient context	—	.	.

Table 6: Kinds of increment, their family, and their role in each task: positive (+), negative (—), unused (.), or labelled by the model’s behaviour ($\pm$). The sufficiency label y^{suff} is 1 exactly for the decisive family.

same index in every row. In the same forward pass we append the gold answer tokens after the anchor. Because attention is causal, this leaves the anchor’s activations unaffected, but it gives the log-probability the model assigns to the gold answer, which we use as a nuisance covariate. The cached keys and values are then cut back to the prompt, and the model decodes greedily from there. Layer index l denotes the residual stream after block l ($l = 0$: embeddings).

Probes. Probes are scikit-learn logistic regressions with L2 regularisation ($C = 0.5$) and balanced class weights, on features standardised on the training split (Alain and Bengio, 2016); the layer is selected by grouped 3-fold cross-validation on training questions (never on test), and the reported number is the test AUROC at that layer. The probe direction is the weight vector w divided by the per-feature standard deviations σ , so that it lives in the original activation space, and normalised to unit norm. The difference-of-means direction is the normalised difference of class means of Δh (or of h for state tasks). Bootstrap intervals resample test questions (1,000 draws) using the stored probe weights.

State probes. State probes use the stored anchor activation of each condition, one example per condition, with the same estimator, standardisation, layer selection and question-level splits as the update probes. For a two-hop question the state-sufficiency task has five positive conditions: the minimal sufficient context in each hop

order (de-duplicated when identical), sufficient plus a distractor, sufficient plus a duplicate, and the full context. It has ten negative ones: no evidence, distractors only, each single hop, and each penultimate state plus a distractor, a duplicate or an answer-string control. Conditions containing a falsified paragraph are excluded. All positives except the minimal sufficient context contain three or more paragraphs, and no negative does, so the task is partly separable by paragraph count; Appendix F reports a count-matched version restricted to two-paragraph contexts. State-correctness and abstention-flag probes use every condition, labelled by the model’s own answer.

Nuisance covariates. Δ tokens, total tokens, Δ gold log-probability, Δ next-token entropy, Δ top probability, answer string in added text / in the full context / in the previous context, TF-IDF cosine relevance of the added passage to the question, number and word count of added paragraphs, relative insertion position, and a 16-dimensional truncated-SVD embedding of the added text’s TF-IDF vector (fit on training increments). The nuisance-only classifier is a logistic regression on these covariates. The residualised probe first regresses Δh on the covariates with ridge regression ($\lambda = 1$) and then probes what the covariates do not explain. The “gain” is the AUROC added when out-of-fold probe scores (5-fold, grouped by question) are given to the nuisance classifier as an extra feature.

Text baselines. TF-IDF (1–2-grams, sublinear tf, $\leq 200k$ features) with logistic regression; DeBERTa-v3-base fine-tuned with AdamW (lr 3×10^{-5} , linear decay, class-weighted cross-entropy). The reduced budget uses one epoch, 256 tokens and 8,000 training increments (all positives kept); the full budget two epochs, 512 tokens and all training increments.

Interventions. Steering adds the vector $\alpha u v$ at the anchor and every generated position (“anchor onward”) or at all positions; the unit u is $|\mathbb{E}[v^\top h \mid \text{sufficient}] - \mathbb{E}[v^\top h \mid \text{insufficient}]|$ on training conditions, computed per layer, so that $\alpha = 1$ moves the activation by one gap between the two class means. Projection-out sets $v^\top h$ to the insufficient-state mean at the anchor and generated positions for every layer from the best layer onward. Patching replaces the anchor residual during the prefill at every layer from the best layer to the last; decoding then proceeds from the patched cache. Principal components are computed from 1,500 training decisive updates per layer; the “unembedding” direction is the normalised row of the output embedding matrix for the gold answer’s first token.

C Readout details: layer depth and projections by increment kind

Layer depth. Every probe in the paper is fitted at every layer, and the main text reports the layer selected by grouped cross-validation on the training questions. Figure 2 shows the full profile. Sufficiency is readable from early layers and peaks at 50–60% of the network’s depth in all three models (layer 17 of 28 for Qwen), after which it declines slowly. Uptake and correction emerge later and peak near the end of the network (layer 26 of 28 for Qwen), which is where the content that determines the answer is assembled. The matched sufficiency task follows the sufficiency profile a few points lower. The same picture holds on the question, answer and relation splits.

Projections by increment kind. Figure 3 projects every held-out increment onto the sufficiency direction, that is, it computes its score along the probe’s weight vector, and averages by kind of increment. The kinds are ordered as one would expect if the direction encoded “the evidence has just become complete”: decisive > falsified > distractor > duplicate > answer-string control > a gold paragraph that does not yet complete

the evidence. Increments added after the context is already sufficient, and the first hop added to an empty context, score far below. That the falsified paragraph scores second indicates that the direction encodes “the final slot has been filled” rather than truth, which is all the model can register before it generates; the behavioural counterpart is that a falsified final paragraph halves abstention (Appendix F).

D Related transitions: correction, stability and revision

Sufficiency and uptake are the two properties the paper is about, but every increment carries the model’s full behavioural transition, and this lets us probe three related properties with the same machinery. *Correction* is defined on increments that start from a wrong or abstaining answer and asks whether the answer becomes correct; unlike uptake it includes non-decisive increments, so a correct answer may also come from guessing a hop or from a falsified paragraph. *Stability* is defined on increments that start from a correct answer and asks whether it stays correct. *Revision* is defined on all increments and asks whether the answer changes at all. Table 7 gives the probe results alongside those of the main text.

Results. Correction is readable at 0.73–0.82 and stability at 0.75–0.87, but both are largely explained by surface properties, chiefly the change in the model’s output distribution: the nuisance-only classifier reaches 0.81–0.85 on correction and 0.90–0.92 on stability, and the probe adds at most 0.06. Revision is readable at 0.71–0.77 (Table 8). The model’s own output is an even stronger predictor of these three: whether the model answers after the increment predicts correction at 0.88–0.90 and stability at 0.80–0.90, because a corrected answer is by definition an answer and a destabilised one is often an abstention. These tasks are therefore less informative about the internal state than uptake, which is the one task on which the activations add far more than surface properties or the output.

Cross-task transfer. Figure 4 applies the direction learned for one task (row) to the increments of another (column), at the row task’s best layer, on the test questions of the document split. Uptake and correction share a direction: the cosine between them is 0.30–0.34, and each predicts the

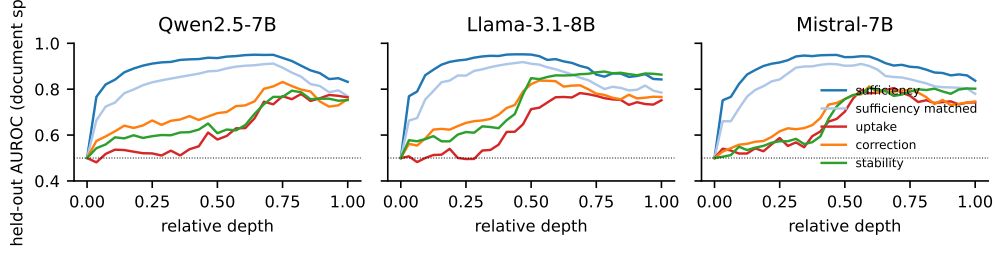


Figure 2: Held-out AUROC by relative depth for the update tasks (document split). Sufficiency peaks around mid-depth in every model; uptake and correction emerge later.

Task	Qwen2.5-7B			Llama-3.1-8B			Mistral-7B		
	AUROC [95% CI]	nuis.	gain	AUROC [95% CI]	nuis.	gain	AUROC [95% CI]	nuis.	gain
Sufficiency	0.948 [0.94,0.95]	0.87	0.11	0.952 [0.95,0.96]	0.86	0.12	0.949 [0.94,0.95]	0.87	0.11
Sufficiency (matched)	0.911 [0.90,0.92]	0.90	0.07	0.918 [0.91,0.93]	0.89	0.08	0.910 [0.90,0.92]	0.90	0.07
Uptake	0.775 [0.74,0.80]	0.66	0.14	0.783 [0.75,0.81]	0.58	0.21	0.771 [0.74,0.80]	0.59	0.18
Correction	0.730 [0.70,0.76]	0.81	0.01	0.817 [0.79,0.84]	0.83	0.06	0.792 [0.77,0.81]	0.85	0.03
Stability	0.750 [0.72,0.78]	0.92	-0.01	0.869 [0.85,0.89]	0.90	0.02	0.800 [0.77,0.83]	0.90	0.01

Table 7: Held-out AUROC of the best-layer probe on paired updates for all five tasks (document split; 95% question-level bootstrap intervals), with the nuisance-only AUROC and the gain from adding the probe score to the nuisance classifier.

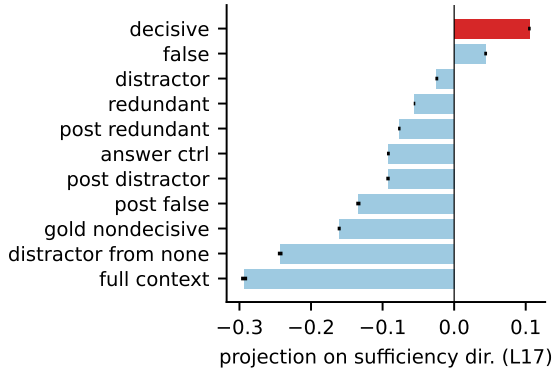


Figure 3: Mean projection of held-out increments on the sufficiency direction, by kind of increment (Qwen2.5-7B, document split; standard errors). Red: decisive.

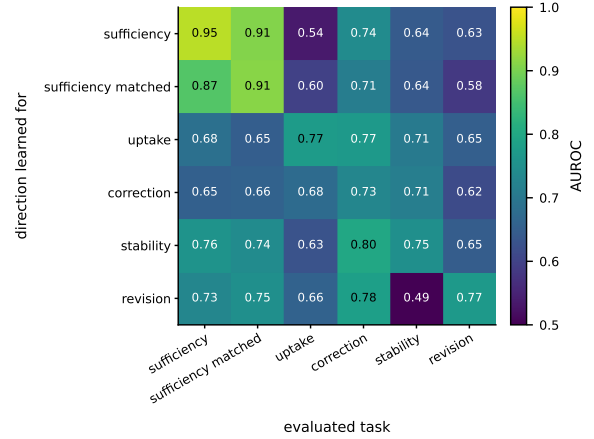


Figure 4: Cross-task transfer (Qwen2.5-7B, document split): AUROC of the direction learned for the row task, evaluated on the column task.

other task at 0.82. Both are nearly orthogonal to sufficiency (cosine 0.02–0.03), whose direction reaches only 0.54–0.60 on uptake. Stability and revision form a third group, aligned with the change in output distribution.

E State versus update: which representation carries the readout?

The probes of Table 2 act on the paired update $\Delta h = h(C_k) - h(C_{k-1})$. A natural question is how much of their accuracy comes from the *change* of state, and how much from information already present in either absolute state. If $h(C_k)$ alone did as well, the finding would be that the

model’s current state represents whether its evidence is sufficient. If $h(C_k)$ were much worse than Δh , sufficiency would be encoded specifically in the transition caused by the new evidence.

Protocol. For every update task we refit the probe on four representations of the *same* increments, with the same labels, document split, regularisation ($C = 0.5$, balanced classes) and layer selection (grouped three-fold cross-validation on training questions, over all blocks):

- $h(C_{k-1})$, the anchor state before the new evidence;

- $h(C_k)$, the anchor state after it;
- Δh , the paper’s representation (it reproduces Table 2 to within 0.002);
- the concatenation $[h(C_{k-1}); h(C_k)]$, from which a linear probe can represent Δh , either state, or any other linear combination of the two.

Both states are the stored activations of the two stateless forward passes; Δh is their difference to within float16 precision.

Sufficiency. The state *after* the new evidence carries most of the readout. $h(C_k)$ reaches 0.930–0.940 on the full task and 0.900–0.908 on the matched one, against 0.947–0.952 and 0.910–0.917 for Δh . The state *before* the evidence is uninformative beyond the trajectory stage. On the full task it scores 0.733 in all three models, which is exactly the AUROC of an oracle that knows only whether the increment starts from the empty context, a partial context or an already sufficient one: 20% of increments starting from a partial context are decisive, and none of the others. On the matched task, where the decisive increment and its four controls start from the same state, it scores exactly 0.500. The concatenation matches Δh (0.949–0.950; matched 0.907–0.923). So subtracting the earlier state adds no information that having both states does not; what it does is remove the large question-specific component that both states share, which makes a single linear direction work.

Why $h(C_k)$ is not trivially enough. On the full task, 27% of negatives end in a context that is already sufficient: a distractor, duplicate or falsified paragraph added after sufficiency, or the full context. A pure “is this context sufficient?” detector cannot reject them. Applying the state-sufficiency probe of Table 2 (trained on whole contexts, 0.99 AUROC on its own task) to $h(C_k)$ of these increments gives only 0.70 on the full update task, and 0.87–0.90 on the matched task, where no negative ends in a sufficient context. The $h(C_k)$ probe can, because the state after the decisive paragraph differs from the state after a post-sufficiency addition: the former is a *minimal* sufficient context, the latter has more paragraphs. Part of what separates them is plausibly context size. On the matched task this cannot help, because every control adds exactly one paragraph to the same penultimate state.

There, $h(C_k)$ at 0.90 is a genuine readout of sufficiency from the resulting state.

Uptake, correction, stability, revision. For the behavioural tasks the state after the evidence is as informative as the update or more so: uptake 0.769–0.798 against 0.771–0.783, correction 0.766–0.872 against 0.730–0.818, stability 0.813–0.882 against 0.750–0.869, revision 0.789–0.806 against 0.709–0.767. The concatenation is best or tied on most of them. The earlier state carries a little information of its own here (0.52–0.65): how the model behaves after new evidence depends partly on the state it was in. These tasks are about the model’s *resulting* answer, so it is unsurprising that the resulting state predicts them best. The uptake result means the signal that decisive evidence will be integrated is present in the state after the evidence arrives, before generation, and does not require comparing with the state before it.

What this means for the paper’s claims.

- The informational claims stand: sufficiency and uptake are linearly readable at the answer-start position before generation, robustly across splits and models. But “readable from the update” should be understood as “readable from the resulting state, measured in a design that controls for the starting state”. The paired update is the cleanest way to read the state change without question-specific content, and it is what makes the matched controls possible (same starting state). It is not the source of the information.
- The causal analysis of Section 6 is unaffected. It patches the anchor residual of the insufficient prompt with that of the sufficient prompt, which is the same as adding Δh to $h(C_{k-1})$. Its conclusion, that the probe direction is inert while the full residual and v_{dm} are causal, concerns directions in this space whichever way they were found.
- The text-baseline comparison (Section 5.1) is unaffected: it compares classifiers on the same increments and labels.

F Behaviour, verbalised sufficiency and the internal state

This appendix does three things. It describes how the models behave on each condition family,

Model	Task	$h(C_{k-1})$	$h(C_k)$	Δh	$[h(C_{k-1}); h(C_k)]$
Qwen2.5-7B	Sufficiency	0.733	0.930	0.947	0.949
	Sufficiency (matched)	0.500	0.901	0.911	0.919
	Uptake	0.605	0.798	0.775	0.780
	Correction	0.574	0.766	0.730	0.782
	Stability	0.549	0.813	0.750	0.832
	Revision	0.556	0.806	0.767	0.791
Llama-3.1-8B	Sufficiency	0.733	0.936	0.952	0.950
	Sufficiency (matched)	0.500	0.908	0.917	0.923
	Uptake	0.570	0.794	0.783	0.788
	Correction	0.522	0.872	0.818	0.883
	Stability	0.592	0.882	0.869	0.888
	Revision	0.563	0.795	0.709	0.751
Mistral-7B	Sufficiency	0.733	0.940	0.948	0.949
	Sufficiency (matched)	0.500	0.900	0.910	0.907
	Uptake	0.647	0.769	0.771	0.775
	Correction	0.566	0.822	0.790	0.819
	Stability	0.524	0.857	0.800	0.857
	Revision	0.634	0.789	0.721	0.790

Table 8: Held-out AUROC (document split, best layer by inner cross-validation) of probes for each update task, fitted on four representations of the same increments. Bold: best per row.

which supplies the behavioural labels behind uptake, correction, stability and revision. It shows that the model’s own output, read as a judgement about whether the evidence is sufficient, is frequently wrong. And it shows that the sufficiency state read from the residual stream at the same position, in the same forward pass, is a much more accurate signal than that verbalised judgement, including in the cases where the verbalised judgement is wrong.

F.1 Behaviour by condition family

Table 9 reports, for every condition family of Appendix A, the fraction of greedy answers that are correct and the fraction that are INSUFFICIENT. It covers the evidence-only regime over the 2,000 two-hop questions of each model. The remaining fraction is a wrong answer. All questions are ones the model answers incorrectly closed-book (Section 3), so a correct answer can be attributed to the evidence.

Read row by row:

- **No evidence and distractors only.** The instruction is followed. The models almost never answer (94–100% INSUFFICIENT) and are never correct. With distractors alone, the 3–6% of prompts that are answered are hallucinations.
- **One hop.** With only one of the two gold paragraphs, the evidence is insufficient by construction, yet the models answer 20–31% of

the time and are correct 10–13% of the time. Some second hops can be guessed from the first, which is one reason correctness and sufficiency are separate labels.

- **Minimal sufficient.** With both gold paragraphs present and nothing else, the models answer correctly only 36–47% of the time, and they say INSUFFICIENT 30–36% of the time. The evidence is sufficient by construction: MuSiQue composes each question from single-hop questions whose supporting paragraphs are exactly these gold paragraphs. An INSUFFICIENT output here is therefore a *verbalised false negative*: the model declares the evidence insufficient when it is not. The remaining 22–28% are wrong answers.
- **One hop plus a falsified final hop.** Replacing the decisive paragraph with a version whose sub-answer is swapped roughly halves abstention (49–55% INSUFFICIENT, against 69–80% with one hop alone). The models treat a type-plausible but false final hop much as they treat the true one, which is what the sufficiency readout also does (Section 5: the false increment projects second only to the decisive one).
- **Sufficient plus a distractor, and the full context.** Adding irrelevant paragraphs to a sufficient context makes the models abstain *more*: 35–47% with one extra distrac-

Condition (correct / abstain)	Qwen2.5-7B	Llama-3.1-8B	Mistral-7B
no evidence	0.00 / 1.00	0.00 / 1.00	0.00 / 0.99
distractors only	0.00 / 0.97	0.00 / 0.97	0.00 / 0.94
one hop	0.10 / 0.80	0.12 / 0.76	0.13 / 0.69
minimal sufficient	0.46 / 0.30	0.47 / 0.31	0.36 / 0.36
one hop + false final hop	0.13 / 0.49	0.13 / 0.55	0.12 / 0.51
sufficient + distractor	0.39 / 0.35	0.38 / 0.41	0.27 / 0.47
full context (20 par.)	0.22 / 0.38	0.17 / 0.61	0.17 / 0.48

Table 9: Behaviour by condition family (evidence-only regime, 2,000 questions per model): fraction correct / fraction answering INSUFFICIENT. The remaining fraction is a wrong answer.

tor and 38–61% with all 20 paragraphs. Correctness falls accordingly (0.17–0.39). Verbalised sufficiency judgements degrade as the context grows, even though the evidence is unchanged.

E.2 The verbalised judgement is a poor sufficiency detector; the internal state is a good one

Because the prompt instructs the model to output INSUFFICIENT exactly when the evidence does not determine the answer, its output is a binary sufficiency judgement: *answers* means “judged sufficient”, *INSUFFICIENT* means “judged insufficient”. We compare this verbalised judgement with the state-sufficiency probe (Section 4), which reads the residual stream at the answer-start anchor of the same forward pass, before any token is generated.

Protocol. The comparison uses the held-out conditions of the document split, with conditions containing falsified evidence excluded, as in the probe task. These are about 2,700 sufficient and 6,000 insufficient contexts per model. For the verbalised judgement we report its true-positive rate (sufficient contexts answered), its false-positive rate (insufficient contexts answered) and its AUROC as a binary score. For the probe we report AUROC and two operating points, both with the threshold set on *training* contexts. The first matches the false-positive rate of the verbalised judgement, so the probe raises false alarms on insufficient contexts as often as the model’s own answers do. The second requires 0.95 precision. At each operating point we report the probe’s recall on the sufficient contexts the model wrongly called INSUFFICIENT (*wrong abstentions*), and its false-alarm rate on the insufficient contexts the model correctly called INSUFFICIENT.

A confound, and how we remove it. Every condition with three or more paragraphs is sufficient (sufficient plus a distractor, sufficient plus a duplicate, full context). Every insufficient condition has at most two. Paragraph count, and with it prompt length, therefore separates part of the state task on its own. This inflates the state probe’s headline AUROC (0.99) and makes its recall on wrong abstentions in the three larger families trivially perfect. We therefore also report a *paragraph-count-matched* version. It keeps only two-paragraph contexts: the minimal sufficient context against distractors only and the penultimate state plus a distractor, a duplicate or an answer-string control. The probe is retrained on these contexts alone, so neither count nor length can separate the classes. The count-matched numbers are the ones we interpret.

Results (Table 10).

- **The verbalised judgement is wrong on a third or more of sufficient contexts.** On held-out data the models call 33% (Qwen), 40% (Llama) and 43% (Mistral) of sufficient contexts INSUFFICIENT, while still answering 15–21% of insufficient ones. As a sufficiency classifier the output reaches 0.76, 0.72 and 0.68 AUROC (0.77, 0.77, 0.71 on two-paragraph contexts).
- **The internal state is far more accurate, with paragraph count held fixed.** On two-paragraph contexts the retrained probe separates minimal sufficient from insufficient contexts at 0.976, 0.981 and 0.986 AUROC. This is 0.20–0.27 above the verbalised judgement made in the same forward pass.
- **It is accurate precisely where the verbalised judgement fails.** Restricted to the two-paragraph contexts the model called INSUFFICIENT, so that the output is held fixed,

	Qwen2.5-7B	Llama-3.1-8B	Mistral-7B
<i>Held-out contexts</i>			
sufficient / insufficient	2754 / 6090	2701 / 6000	2754 / 6100
sufficient but verbalised INSUFFICIENT	901 (33%)	1079 (40%)	1183 (43%)
<i>Verbalised judgement (answers = sufficient)</i>			
AUROC	0.762	0.722	0.681
TPR / FPR	0.673 / 0.150	0.601 / 0.156	0.570 / 0.208
<i>State-sufficiency probe at the same anchor</i>			
AUROC	0.992	0.994	0.996
TPR / FPR, threshold at verbalised FPR	0.995 / 0.188	0.996 / 0.178	1.000 / 0.194
recall on wrongly abstained sufficient contexts	0.990	0.991	0.999
false alarms on correctly abstained insufficient contexts	0.154	0.151	0.167
TPR / FPR, threshold at 0.95 precision	0.962 / 0.050	0.970 / 0.051	0.977 / 0.038
recall on wrongly abstained sufficient contexts	0.936	0.956	0.961
false alarms on correctly abstained insufficient contexts	0.039	0.040	0.029
<i>Wrong abstentions by condition: abstain rate; probe recall at 0.95 precision</i>			
minimal sufficient	0.29; 0.79	0.30; 0.84	0.37; 0.87
sufficient + distractor	0.35; 1.00	0.41; 1.00	0.49; 1.00
sufficient + duplicate	0.29; 0.99	0.34; 0.99	0.39; 1.00
full context (20 par.)	0.40; 1.00	0.60; 1.00	0.50; 1.00
<i>Mean probe score (0 = insufficient mean, 1 = sufficient mean on training contexts)</i>			
sufficient, answered	0.98	0.88	0.95
sufficient, verbalised INSUFFICIENT	1.03	1.15	1.04
insufficient, answered	0.10	0.08	0.04
insufficient, verbalised INSUFFICIENT	-0.01	-0.00	-0.03
<i>Two-paragraph contexts only (paragraph count matched; probe retrained on them)</i>			
minimal sufficient / insufficient two-paragraph	927 / 4263	901 / 4200	924 / 4270
minimal sufficient but verbalised INSUFFICIENT	268	273	341
verbalised judgement: AUROC; TPR / FPR	0.774; 0.71 / 0.16	0.773; 0.70 / 0.15	0.711; 0.63 / 0.21
probe AUROC (all two-paragraph contexts)	0.976	0.981	0.986
probe AUROC among contexts verbalised INSUFFICIENT	0.967	0.972	0.982
threshold at verbalised FPR: TPR / FPR	0.984 / 0.196	0.986 / 0.187	0.997 / 0.213
recall on wrong abstentions; false alarms on correct abstentions	0.963; 0.171	0.963; 0.161	0.991; 0.188
threshold at 0.95 precision: TPR / FPR	0.859 / 0.041	0.890 / 0.047	0.905 / 0.039
recall on wrong abstentions; false alarms on correct abstentions	0.757; 0.032	0.773; 0.037	0.812; 0.028
mean probe score: sufficient answered / sufficient abstained	1.03 / 0.82	1.03 / 0.80	1.03 / 0.87
mean probe score: insufficient answered / insufficient abstained	0.16 / -0.01	0.14 / -0.01	0.08 / -0.04

Table 10: Verbalised versus internal sufficiency judgements on held-out contexts (document split; falsified-evidence contexts excluded). Verbalised: the model answers (judged sufficient) or says INSUFFICIENT. Probe: state-sufficiency probe at the answer-start anchor of the same forward pass; thresholds are set on training contexts. “Wrong abstentions”: sufficient contexts the model called INSUFFICIENT; “correct abstentions”: insufficient contexts it called INSUFFICIENT. Mean probe scores are expressed so that 0 is the mean of insufficient and 1 the mean of sufficient training contexts. Top block: all contexts, inflated by paragraph count. Bottom block: two-paragraph contexts only, with the probe retrained on them.

the probe still separates the sufficient ones from the insufficient ones at 0.967–0.982. At 0.95 training precision it recovers 76%, 77% and 81% of the wrongly rejected minimal sufficient contexts, with 3–4% false alarms on correctly rejected insufficient ones. At the verbalised false-alarm rate it recovers 96–99% of them. In practical terms, a detector reading the anchor before generation would accept most of the complete evidence the model itself refuses, at a fraction of the false alarms the model’s own answers produce.

when rejected and 0.08–0.16 when (wrongly) answered. The internal sufficiency state is graded. The verbal decision behaves like a thresholded and noisier read-out of it, which tends to fail when the state is weaker than usual, but the state in those cases is still much closer to “sufficient” than to “insufficient”. (Without the count matching, the wrongly rejected contexts appear to score as high as answered ones, 1.03–1.15. That impression comes from the larger, trivially separable families and should not be relied on.)

- **But the internal state is weaker in exactly those cases.** In units where the insufficient training mean is 0 and the sufficient one is 1, wrongly rejected minimal sufficient contexts score 0.82, 0.80 and 0.87, against 1.03 for sufficient contexts the model answers. Insufficient contexts score about 0

- **Adding paragraphs hurts the verbal judgement.** The abstention rate on sufficient contexts rises from 29–37% (minimal) to 40–60% (full context), while every wrongly rejected full-context prompt is still on the sufficient side of the probe’s threshold. Paragraph count makes that last comparison easy, so we

do not interpret it further.

F.3 Holding the output fixed for the update probes

The same question can be asked of the update probes of Table 2: is “the evidence just became sufficient” anything more than “the model just switched from INSUFFICIENT to an answer”? Table 11 reports, for the sufficiency and matched-sufficiency tasks with Δh and with $h(C_k)$ (Appendix E), and for the state-sufficiency task, four numbers:

- the AUROC of the output alone: the indicator that the model answers at C_k (for updates, the indicator that it switched from INSUFFICIENT to an answer scores within 0.04 of this);
- the standard probe, trained on all increments, scored separately on the test increments after which the model *still says INSUFFICIENT* and on those after which it answers (“stratified”);
- probes trained *and* tested only within each of these two groups (“within”). Within the abstaining group, the positives are decisive increments that completed the evidence while the model went on refusing, and the negatives are controls after which it also refused.

The output alone gives 0.65–0.76. The probes lose almost nothing when the output is held fixed. Among increments after which the model still says INSUFFICIENT, the Δh sufficiency probe scores 0.940–0.949 (stratified) and 0.935–0.944 (within), against 0.945–0.950 overall; for $h(C_k)$ the corresponding figures are 0.930–0.944 and 0.914–0.933. Among increments after which the model answers, the probes score slightly lower (0.929–0.942 for Δh), because there the controls that the model answers (false evidence, answer-string controls) are the hardest ones. On the matched task, where every control starts from the same state as the decisive increment, the probes reach 0.883–0.887 even when trained and tested only on increments the model refuses.

F.4 What this does and does not show

- **The sufficiency readout is not the planned output.** If the probes were reading “about to output INSUFFICIENT”, they would fall to

chance once the output is held fixed. They stay at 0.88–0.99 depending on the task. The residual stream at the answer-start position encodes that the context contains the complete evidence, separately from the decision to answer.

- **The internal signal is a better sufficiency detector than the model’s self-report.** For a system that has to decide whether retrieved evidence is complete, the anchor activation, read before generation, is more accurate than asking the model and reading its answer: 0.98 against 0.71–0.77 AUROC with paragraph count matched. At 0.95 precision the probe accepts 86–91% of sufficient contexts with 4% false alarms, whereas the model’s answers accept 63–71% with 15–21% false alarms.
- **This is a statement about evidence completeness, not about the answer.** The probe is trained on construction labels (all gold hops present). A high score on a wrongly rejected context says that the representation registers the evidence as complete. It does not say that the model could compute the answer from it. Whether it will is what uptake measures (Section 5), and uptake is lower and nearly orthogonal to sufficiency. Among the minimal sufficient contexts it rejects, the model may lack the bridging step, and the internal sufficiency state cannot supply it. Consistently, using the sufficiency score to stop retrieval does not by itself improve accuracy (Appendix R): stopping at a sufficient context the model will not answer from still yields INSUFFICIENT.
- **The causal picture is consistent with this division.** The one causally active direction, v_{dm} , acts as an abstention gate (Section 6): pushing along it makes the model answer, including on some sufficient contexts it was rejecting (positive steering adds 3 points of correct answers on Qwen and 18 on Llama), but also on distractors and false evidence. The sufficiency *readout* is not that gate: projecting the probe direction out changes nothing. The model represents completeness accurately along directions it does not use for the answer/abstain decision, and its decision is driven by a lower-dimensional, noisier

Model	Task	Repr.	output	all	model abstains at C_k		model answers at C_k	
			only	test	stratified	within	stratified	within
Qwen2.5-7B	Sufficiency	Δh	0.685	0.945	0.948	0.944	0.929	0.927
	Sufficiency	$h(C_k)$	0.685	0.930	0.930	0.918	0.904	0.890
	Matched	Δh	0.721	0.911	0.891	0.884	0.879	0.870
	Matched	$h(C_k)$	0.721	0.909	0.904	0.883	0.854	0.843
	State suff.	$h(C)$	0.762	0.992	0.990	0.982	0.986	0.983
Llama-3.1-8B	Sufficiency	Δh	0.692	0.950	0.940	0.935	0.942	0.929
	Sufficiency	$h(C_k)$	0.692	0.927	0.931	0.914	0.893	0.892
	Matched	Δh	0.726	0.912	0.896	0.883	0.875	0.855
	Matched	$h(C_k)$	0.726	0.908	0.908	0.886	0.840	0.838
	State suff.	$h(C)$	0.722	0.992	0.992	0.988	0.986	0.982
Mistral-7B	Sufficiency	Δh	0.646	0.948	0.949	0.937	0.939	0.931
	Sufficiency	$h(C_k)$	0.646	0.936	0.944	0.933	0.912	0.912
	Matched	Δh	0.671	0.910	0.918	0.887	0.874	0.876
	Matched	$h(C_k)$	0.671	0.902	0.909	0.872	0.860	0.856
	State suff.	$h(C)$	0.681	0.996	0.996	0.994	0.994	0.989

Table 11: Sufficiency probes with the model’s output held fixed (document split). Output only: AUROC of the indicator “the model answers at C_k ”. All: standard probe, all test increments. Stratified: the same probe on the test increments after which the model abstains (resp. answers). Within: probe trained and tested only on that group, layer re-selected by grouped inner cross-validation. The state-sufficiency rows are not count-matched (see Table 10 for the count-matched version).

gate.

- **Scope.** The comparison uses MuSiQue two-hop questions the models answer incorrectly closed-book, under the evidence-only instruction. With a more permissive instruction (Appendix H), the models abstain less on sufficient evidence and follow false evidence more, so the verbalised false-negative rate depends on the prompt; the internal update readout barely changes (0.945 against 0.948).

F.5 Uptake and the decision to answer

The same question can be asked of the uptake probe: how much of its 0.78 AUROC is the model’s own decision to answer? Uptake is positive when the decisive increment makes the answer correct, and a correct answer is by definition an answer, whereas 45–55% of the decisive increments the model fails to take up are ones after which it says INSUFFICIENT. The indicator “the model answers after the increment” therefore reaches 0.78 on its own, exactly the probe’s AUROC (Table 3). Table 12 restricts the comparison to the decisive increments the model answers, where the only remaining question is whether the answer is right. Three internal or output signals are compared there: the output confidence (negative next-token entropy at the anchor; top-token probability gives the same numbers to within 0.005), the stored uptake probe trained on all decisive increments, and a probe refitted on the answered increments only, at the stored best layer. The right-hand

block combines each signal with the decision to answer, ranking abstentions below every answer and answers by the signal.

Among answered increments the probe separates correct from wrong answers at 0.63–0.68 and confidence at 0.63–0.64: the probe is ahead by 0.03 on Qwen and 0.05 on Llama and level on Mistral. Combined with the decision to answer, the probe reaches 0.84–0.86 and confidence 0.84. Two things follow. The uptake probe does carry information that the text does not have (no text classifier exceeds 0.60) and that the model’s output does not fully have, but the larger part of its 0.78 is the decision to answer, which the model verbalises for free. And the part specific to correctness is modest and only a little better than the model’s own confidence, so claims that internal states predict evidence uptake should be read with this decomposition in mind. The verification experiment of Appendix R, which uses the uptake direction to decide whether to trust the answer given once retrieval stops, measures the same quantity on a different set of contexts and reaches the same conclusion.

G Robustness across data splits

Every probe is trained on 70% of questions and tested on the remaining 30% under four groupings that are never shared between train and test: question id; gold document, where questions that share any supporting page are grouped together (transitively) and each group goes entirely to one side; answer entity; and the tuple of relation types along

Model	n	correct	answered increments only			all decisive increments		
			confidence	probe	probe (refit)	verbalised	+ confidence	+ probe
Qwen2.5-7B	867	0.67	0.636	0.658	0.665	0.776	0.837	0.850
Llama-3.1-8B	833	0.66	0.632	0.651	0.678	0.783	0.840	0.860
Mistral-7B	778	0.53	0.638	0.629	0.638	0.775	0.837	0.837

Table 12: Uptake among the decisive increments the model answers (document split, test questions). n : answered test increments; “correct”: fraction correct among them. Left: AUROC for correct against wrong answers of output confidence, of the uptake probe trained on all decisive increments, and of a probe refitted on answered increments. Right: AUROC on all decisive increments of the decision to answer alone and combined with confidence or with the refitted probe.

the question’s hops. The document split is the default in the main text. Table 13 gives the best-layer test AUROC under all four for the three two-hop runs, the 3–4-hop run and 2WikiMultiHopQA. Sufficiency varies by at most 0.015 across splits. Uptake is lower on the document and answer splits than on the question split, by 0.06–0.13 on the two-hop runs and by 0.18 on 3–4-hop questions (Appendix K).

H Evidence-augmented regime

Setup. The evidence-augmented regime repeats every stateless forward pass of the Qwen two-hop run with one change to the instruction: “use the supplied evidence together with your own knowledge” instead of “use only the supplied evidence”. Conditions, increments, splits and probes are otherwise identical.

Results. The Qwen probes give sufficiency 0.945, matched 0.913, uptake 0.721 and correction 0.786 (document split), within 0.05 of the evidence-only regime. Behaviour shifts as expected for a more permissive instruction: on minimally sufficient contexts the model answers more (0.51 correct, 0.23 abstain) and it follows false evidence more often.

I Transfer without retraining

Table 14 applies the Qwen 2-hop stateless directions, unchanged and at the layer chosen on MuSiQue, to the conversational protocol (Appendix J), to 2WikiMultiHopQA (Appendix L) and to 3–4-hop MuSiQue (Appendix K). All target increments are scored, since none of them were used to train the source probe. Transfer to the external benchmarks and to retrieved evidence is reported in Section 7 and Appendix T; cross-task transfer between the five tasks is in Appendix D.

J Conversational protocol

Setup. In the conversational revision protocol the model sees, in a two-turn chat, the question with its earlier evidence and its own stateless answer A_k from that context, and then only the new paragraph, with the instruction to change its answer only if the total evidence warrants it. The paired update is $h_{\text{conv}}(C_k \rightarrow C_{k+1}) - h(C_k)$. The 10 chain-related increment kinds are re-run in this format (30,000 increments for Qwen); labels are defined as in the stateless protocol.

Results. Probes trained within the conversational regime reach sufficiency 0.951, uptake 0.818, correction 0.922, stability 0.959 and revision 0.969 (document split). Stability and revision are much higher than in the stateless regime because the model’s previous answer is in the prompt. The stateless directions transfer at 0.79 (sufficiency), 0.78 (uptake), 0.92 (correction) and 0.82 (stability) (Table 14).

K 3–4-hop questions

Setup. A separate Qwen run uses 3- and 4-hop MuSiQue training questions (closed-book accuracy 1.3%), with the condition set of Appendix A including random hop orders and leave-one-out contexts, which exist only for $n \geq 3$.

Results. Sufficiency probes reach 0.931/0.925/0.923/0.932 (question/document/answer/relation split); matched 0.907/0.900/0.896/0.906; uptake 0.795/0.613/0.611/0.795. Along the sufficiency direction, chain-decisive and leave-one-out-decisive increments have the same mean projection (+0.023), followed by false (+0.003), distractor (−0.02), redundant and answer-string (−0.06) and non-decisive gold increments (−0.11). Patching (correct/answered): none

Run	Task	question	document	answer	relation
Qwen2.5-7B	Suff.	0.956	0.948	0.945	0.947
Qwen2.5-7B	Uptake	0.851	0.775	0.726	0.841
Llama-3.1-8B	Suff.	0.960	0.952	0.958	0.959
Llama-3.1-8B	Uptake	0.845	0.783	0.736	0.815
Mistral-7B	Suff.	0.964	0.949	0.954	0.958
Mistral-7B	Uptake	0.873	0.771	0.758	0.793
Qwen 3-4 hop	Suff.	0.931	0.925	0.923	0.932
Qwen 3-4 hop	Uptake	0.795	0.613	0.611	0.795
Qwen 2Wiki	Suff.	0.986	0.986	0.981	0.983
Qwen 2Wiki	Uptake	0.894	0.894	0.866	0.786

Table 13: Best-layer AUROC under the four disjoint splits, for the three 2-hop runs, the 3-4-hop run and 2Wiki-MultiHopQA.

Direction (MuSiQue 2-hop, stateless)	→ conversational	→ 2WikiMultiHopQA	→ MuSiQue 3-4 hop
sufficiency	0.790	0.956	0.822
sufficiency matched	0.776	0.931	0.849
uptake	0.776	0.791	0.675
correction	0.920	0.846	0.744
stability	0.820	0.822	0.755
revision	0.659	0.770	0.708

Table 14: Transfer of Qwen2.5-7B 2-hop directions without retraining (AUROC at the source’s best layer).

0.05/0.20, full 0.25/0.62, parallel 0.06/0.19, orthogonal 0.25/0.62, PC256 0.20/0.51, random-256 0.06/0.20; projecting out v_{dm} leaves 0.03 answered. The two-hop directions transfer at 0.82–0.85 for sufficiency (Table 14).

L 2WikiMultiHopQA

Setup. 2WikiMultiHopQA (Ho et al., 2020) was built from Wikipedia and Wikidata to evaluate multi-hop reasoning with explicit reasoning paths: each question comes with the evidence triples that connect it to the answer. Because it is constructed independently of MuSiQue, we use it to test whether the probes transfer to a different construction. The Qwen pipeline is run unchanged on its two-hop questions, with gold paragraphs from its supporting facts and its own distractors.

Results. Within-dataset probes (document split): sufficiency 0.986, matched 0.972, uptake 0.894, correction 0.914. Here the matched task is almost fully explained by nuisance covariates (0.965, chiefly the change in gold-answer log-probability); the residualised probe reaches 0.82. The MuSiQue directions transfer at 0.96 (sufficiency), 0.93 (matched), 0.85 (correction) and 0.79 (uptake) (Table 14).

M Closed-book transfer

Setup. This experiment asks whether the directions learned on evidence trajectories also say something when no evidence is given at all. We re-run the closed-book prompt on Qwen’s screened pool of 4,000 questions with activation capture and score every stored direction on the closed-book anchor activations, for two targets: closed-book correctness and closed-book abstention (INSUFFICIENT). A probe trained directly on closed-book activations (question-disjoint split) gives the ceiling.

Results. Closed-book answers are correct 1.5% of the time and abstain 91.7% of the time. For correctness, the state-correctness direction reaches 0.810 (0.838 at its best layer), correction 0.799 and state-sufficiency 0.722 (0.808), against an own-probe ceiling of 0.874. The abstention-flag direction predicts abstention at 0.977, against a ceiling of 0.973.

N Steering along the logistic directions

Setup. We add αuv along the logistic state-sufficiency direction at its best layer, from the anchor onward, with u the sufficient-minus-insufficient gap along v (Appendix B). Doses are $\alpha \in \{\pm 0.5, \pm 1, \pm 2\}$ gap units ($\leq 1.4 \ell_2$) and $\alpha \in \{\pm 8, \pm 16, 32\}$ (5–22 ℓ_2 , on a mean residual norm of 61). We also steer along the uptake direc-

tion at layer 26 ($\leq 16 \ell_2$), and run an all-positions variant. A random unit direction with the same ℓ_2 magnitude is the control. Outcomes are measured on 150 held-out questions in each of five sets: sufficient, insufficient, false-evidence, answer-string and distractor-only contexts.

Results. Every rate stays within ± 0.02 of baseline on all five sets, for the probe directions and the random control alike. At 32 units all rates degrade by 0.02–0.05, again equally for probe and random directions. The all-positions variant at $\leq 11 \ell_2$ is equally flat.

O Difference-of-means steering

Low dose (main protocol). We add $\alpha u_l v_{dm}^{(l)}$ at four layers from the best sufficiency layer, from the anchor onward, with $\alpha \in \{\pm 0.5, \pm 1\}$ in units of the per-layer sufficient-minus-insufficient gap, on 150 held-out questions per set. Table 15 gives the full numbers behind Fig. 1.

High dose. Adding $\alpha u_l v_{dm}^{(l)}$ at all positions and at every layer from the best layer onward with $|\alpha| \geq 2$ (per-layer gaps of 5–14 ℓ_2) produces degenerate output: every prompt is “answered”, never correctly and never with INSUFFICIENT. This happens equally for the probe direction and the random control, and is why the main protocol uses four layers, anchor onward and $|\alpha| \leq 1$.

P How many dimensions the causal content needs

Section 6 shows that patching the whole anchor residual of the insufficient prompt with the sufficient prompt’s makes the model answer, while the component along the probe direction does nothing. To measure how many dimensions the causal content occupies, Fig. 5 patches only the part of the sufficient-minus-insufficient update that lies in the top- k principal components of the decisive updates seen in training, for $k \in \{1, 4, 16, 64, 256\}$, and compares with a random 256-dimensional subspace, with the gold answer’s unembedding direction alone, and with everything but that direction.

The answering rate rises gradually with k (Qwen: 0.29 at $k=1$, 0.49 at $k=16$, 0.59 at $k=256$, 0.71 for the full update), while the random 256-dimensional subspace does nothing. Correctness needs even more of the update than answering does. The first component carries much of the switch between answering and abstaining, which

is consistent with the projection-out result for v_{dm} . What carries correctness differs by model. On Llama the gold answer’s unembedding direction alone restores 0.19/0.62 (correct/answered), most of the full effect. On Qwen and Mistral it restores little (0.09/0.19, 0.07/0.19), although removing it from the full patch halves Qwen’s correctness (0.26 $\rightarrow$ 0.13). Mistral’s and the 3–4-hop run’s effects are captured by the top principal components instead (0.56 and 0.51 answered at $k=256$). The causal content is thus high-dimensional and model-specific, and Appendix Q shows that it is concentrated in the later layers.

Q Single-layer patching

Setup. As in Section 6, the insufficient prompt’s anchor residual is replaced by the sufficient run’s (full patch), but at a single layer only instead of every layer from the best one onward.

Results. Patching only the best sufficiency layer (16 for Qwen) leaves the insufficient prompt unchanged in all variants. Patching only layer 26 with the full sufficient residual restores 0.18/0.62 (correct/answered), so late-layer anchor content alone suffices to make the model answer.

R Adaptive retrieval

The readouts of Section 5 are computed before generation, so they can drive retrieval: after each round, read the anchor and decide whether to stop and answer or to retrieve more. We test every learned direction in this role, both as a *stopping rule* and as a *verifier* of the answer given at the stopping round.

Simulation. Each held-out question is walked along its canonical trajectory, one retrieval round per stage: (1) the matched distractors, (2) the first gold hop, (3) the minimal sufficient context, (4) the sufficient context plus a distractor. A policy that never stops answers from stage 4; an oracle stops at stage 3. At round k a policy reads either the state $h(C_k)$ (state probes) or the update the round produced, $\Delta h = h(C_k) - h(C_{prev})$ (update probes), projects it on the direction at the direction’s best layer, and stops when the projection crosses a threshold. The threshold *and the sign* are chosen on training questions to maximise accuracy at the stopping stage (ties broken by fewer rounds), so each direction is used in whichever

Model	α	sufficient corr./ans.	insufficient ans.	distractors ans.	false ans./followed	answer-string ans.
Qwen2.5-7B	-1.0	0.23/0.34	0.05	0.00	0.23/0.06	0.03
	-0.5	0.34/0.52	0.09	0.01	0.43/0.13	0.09
	+0.5	0.49/0.79	0.20	0.06	0.75/0.25	0.29
	+1.0	0.49/0.82	0.26	0.12	0.79/0.25	0.35
Llama-3.1-8B	-1.0	0.03/0.07	0.00	0.00	0.03/0.01	0.00
	-0.5	0.25/0.38	0.07	0.00	0.23/0.12	0.05
	+0.5	0.65/0.89	0.27	0.13	0.82/0.35	0.34
	+1.0	0.69/0.94	0.51	0.29	0.92/0.39	0.59
Mistral-7B	-1.0	0.01/0.08	0.06	0.11	0.06/0.01	0.04
	-0.5	0.11/0.33	0.09	0.01	0.25/0.09	0.04
	+0.5	0.47/0.78	0.31	0.12	0.75/0.29	0.29
	+1.0	0.54/0.87	0.47	0.28	0.87/0.28	0.43

Table 15: Low-dose steering along v_{dm} (150 held-out questions per set). Baselines ($\alpha = 0$) appear in Fig. 1.

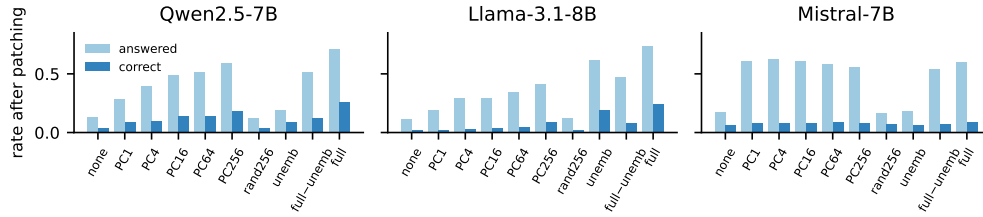


Figure 5: The insufficient prompt patched with the sufficient-minus-insufficient update projected on the top- k principal components of training updates, on a random 256-dimensional subspace, on the gold answer’s unembedding direction (“unemb”), or on everything but that direction.

reading serves retrieval best. For stability and revision we also report the reading that stops when the probe says the last round *changed the answer*. Baselines: never stop; the oracle; the behavioural rule, which stops at the first round whose greedy answer is not INSUFFICIENT (it needs a generation per round, which the probes do not); and a next-token-entropy threshold calibrated the same way. Accuracy counts abstentions as wrong. Everything is computed from stored activations and behaviour, with no new forward passes. Thresholds are calibrated on the questions the probes were trained on, which makes the probe policies, if anything, slightly over-confident on test questions. The trajectories present gold hops in order, so the simulation measures *when to stop*, not what to retrieve.

Stopping rules (Table 16, Fig. 6). Only the sufficiency signals are useful stopping rules. The update-sufficiency v_{dm} direction is the best learned rule on all three MuSiQue models (0.478, 0.468, 0.336). It matches the oracle (0.475, 0.470, 0.336) in about three rounds, and on Qwen it is the one rule significantly better than the behavioural rule. The logistic state and update sufficiency probes are statistically tied with the behavioural

rule on MuSiQue, while acting before any generation. On HotpotQA both beat it (0.732 and 0.728 vs 0.650), and on the synthetic data the state probe does (0.977 vs 0.939), because the behavioural rule stops too early there (HotpotQA: 2.53 rounds). On 2WikiMultiHopQA they tie it (0.583 vs 0.593). The other directions are weak stopping rules on every source. Uptake, correction and stability are trained on decisive or already-correct increments and say nothing about whether the evidence is complete. Their best thresholds stay within 0.06 of the never-stop policy on MuSiQue (0.24–0.42) and below the best sufficiency rule on every source. Read as “stop when the answer changes”, stability never fires (about 4.0 rounds, at full-context accuracy). Revision, the direct change detector, stops at about three rounds but often on the first wrong answer, and is the worst learned rule on MuSiQue (0.236–0.396). The v_{dm} direction of the *state*, which the causal analysis identified as an abstention gate (Section 6), is a poor stopping rule (0.26–0.41 on MuSiQue). Its update counterpart is the best one. Figure 6 sweeps each threshold on test questions: the sufficiency curves peak at the oracle’s operating point, while the uptake and stability curves rise roughly monotonically towards the full-context accuracy and never cross the be-

Stopping rule	MuSiQue Qwen	MuSiQue Llama	MuSiQue Mistral	HotpotQA Qwen	2WikiMultiHopQA Qwen	Synthetic Qwen
<i>Baselines</i>						
full context (never stop)	0.404 (4.00)	0.385 (4.00)	0.243 (4.00)	0.685 (3.99) ⁺	0.460 (4.00)	0.966 (3.65) ⁺
oracle (stop when sufficient)	0.475 (3.00)	0.470 (3.00)	0.336 (3.00)	0.740 (3.00) ⁺	0.583 (3.00)	0.973 (2.93) ⁺
behavioural (first non-INSUFFICIENT)	0.460 (3.09)	0.460 (3.09)	0.318 (3.08)	0.650 (2.53)	0.593 (3.10)	0.939 (2.88)
next-token entropy	0.381 (3.85) [−]	0.378 (3.96)	0.261 (3.91)	0.690 (3.74) ⁺	0.467 (3.67) [−]	0.969 (3.35) ⁺
<i>State probes, $h(C_k)$</i>						
sufficiency (logistic)	0.466 (3.13)	0.448 (3.11)	0.326 (3.13)	0.732 (3.08)⁺	0.583 (3.02)	0.977 (2.96)⁺
sufficiency (v_{dm})	0.407 (3.94)	0.385 (3.58)	0.261 (3.63)	0.690 (3.86) ⁺	0.463 (3.71)	0.967 (3.00) ⁺
correctness	0.399 (3.57)	0.443 (3.51)	0.284 (3.64)	0.667 (2.92)	0.547 (3.42)	0.964 (3.23) ⁺
abstention flag	0.435 (3.07) [−]	0.435 (3.12) [−]	0.302 (3.13) [−]	0.702 (3.18) [−]	0.583 (3.12) [−]	0.969 (2.99) [−]
<i>Update probes, Δh of the round</i>						
sufficiency (logistic)	0.458 (3.30)	0.460 (3.09)	0.316 (3.18)	0.728 (3.23) ⁺	0.570 (3.16)	0.949 (3.32)
sufficiency (v_{dm})	0.478 (3.06)⁺	0.468 (2.98)	0.336 (3.00)	0.727 (3.07) ⁺	0.543 (3.32)	0.970 (3.15) ⁺
sufficiency matched	0.404 (4.00)	0.385 (4.00) [−]	0.243 (4.00)	0.683 (3.98) ⁺	0.513 (3.21)	0.858 (3.22)
uptake	0.417 (3.55)	0.388 (3.35)	0.259 (3.47)	0.683 (3.98)	0.507 (3.42)	0.957 (3.07) ⁺
correction	0.404 (3.43)	0.405 (3.38)	0.302 (3.53)	0.667 (2.93)	0.550 (3.36)	0.920 (3.19)
stability	0.407 (3.46)	0.400 (3.42)	0.243 (4.00) [−]	0.685 (3.99) ⁺	0.543 (3.38)	0.964 (3.63) ⁺
stability, stop when the answer changes	0.402 (3.99)	0.385 (4.00)	0.243 (4.00)	0.685 (3.99)	0.457 (3.99)	0.966 (3.65)
revision (stop when the answer changes)	0.396 (3.13)	0.363 (3.12)	0.236 (3.23)	0.690 (3.90) ⁺	0.513 (3.07)	0.934 (3.02)
test questions	609	600	610	600	300	830

Table 16: Every direction as a stopping rule: accuracy at the stopping round (mean rounds in parentheses), held-out questions of the document split. Bold: best non-oracle rule per column. [−]: the training-optimal reading stops when the projection is *low*. ⁺: the 95% paired bootstrap interval of the accuracy difference to the behavioural rule lies above zero (1,000 resamples of test questions).

havioural rule on MuSiQue.

Verification at the stopping round (Table 17).

The directions that fail as stopping rules are useful once retrieval has stopped. We take the stopping round chosen by the state-sufficiency rule and ask, among the questions the model answers there, how well each signal ranks correct above wrong answers. This is the decision to output, flag or re-retrieve. The uptake direction is the best learned verifier on MuSiQue Qwen (0.672) and 2WikiMultiHopQA (0.847), where it beats output confidence (top-token probability 0.621, 0.705). Correction is the best learned verifier on Mistral (0.666 vs 0.630). By AUROC, next-token entropy is ahead on Llama and HotpotQA (by 0.02) and on the synthetic data (by 0.11, where 99% of answered stops are correct). For selective answering, keeping only the more confident half of the answers, uptake gives higher accuracy than top-token probability in five of six settings (0.789 vs 0.751 on MuSiQue Qwen, 0.957 vs 0.870 on 2WikiMultiHopQA), and is level on the synthetic data (0.995 vs 0.998). The sufficiency directions are near chance as verifiers on MuSiQue and HotpotQA (0.46–0.60), and the abstention flag is below chance everywhere. They detect completeness, which the stopping rule has already selected for, not whether the answer is right. This division of labour mirrors the readout results. Sufficiency tells the system when the evidence is complete. Uptake, nearly orthogonal to it (Section 5), tells

whether the model has actually integrated that evidence into its answer.

Other sources. On 3–4-hop MuSiQue the trajectories end at the sufficient context (no post-sufficiency stage), so the oracle and the never-stop policy coincide (0.360). All learned rules reach this value, while the behavioural rule stops too early (0.340). On RAGBench (relation split, 955 test questions) most trajectories have two or three stages and no distractor after sufficiency, and every policy lies within 0.21–0.23. With the original protocol of a threshold at 0.97 training precision, the state-sufficiency rule reached 0.442 in 2.87 rounds on MuSiQue Qwen and 0.707 in 2.90 on HotpotQA. Accuracy-optimal calibration trades a few rounds for accuracy on MuSiQue.

S Results by question group on the external sources

Table 18 breaks down the held-out probe AUROC of the external sources (Section 7) by question group, to separate the evidence-state readout from properties of the questions. The groups are: closed-book status (unlike the main runs, HotpotQA and RAGBench keep the questions the model answers correctly without evidence); question type (HotpotQA bridge versus comparison, whose two facts are independent lookups); domain (RAGBench’s twelve sub-datasets grouped into biomedical, finance, technical, customer support, general web QA and Wikipedia); and hop

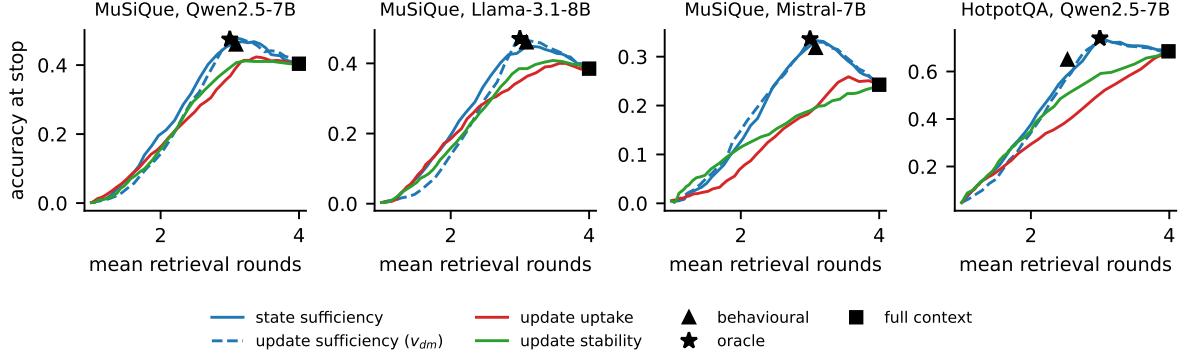


Figure 6: Accuracy against mean retrieval rounds when each direction’s stopping threshold is swept over test questions (41 quantiles), with the behavioural rule, oracle and full context as points.

Signal at the stopping round	MuSiQue Qwen	MuSiQue Llama	MuSiQue Mistral	HotpotQA Qwen	2WikiMultiHopQA Qwen	Synthetic Qwen
top-token probability	0.621	0.615	0.630	0.654	0.705	0.876
—next-token entropy	0.623	0.630	0.637	0.657	0.705	0.877
state: sufficiency	0.465	0.463	0.521	0.526	0.621	0.738
state: correctness	0.560	0.568	0.618	0.626	0.763	0.388
state: abstention flag	0.432	0.473	0.412	0.395	0.444	0.252
update: sufficiency	0.482	0.543	0.495	0.554	0.594	0.585
update: sufficiency (v_{dm})	0.566	0.569	0.601	0.567	0.806	0.534
update: uptake	0.672	0.613	0.647	0.617	0.847	0.764
update: correction	0.594	0.576	0.666	0.639	0.753	0.400
update: stability	0.549	0.566	0.559	0.562	0.775	0.497
update: revision	0.512	0.517	0.490	0.524	0.601	0.531
accuracy of all answered stops	0.665	0.666	0.534	0.810	0.761	0.990
accuracy, top half by top-token prob.	0.751	0.733	0.618	0.863	0.870	0.998
accuracy, top half by uptake	0.789	0.757	0.651	0.867	0.957	0.995
answered stops	427	404	373	542	230	819

Table 17: Verification once retrieval stops (state-sufficiency stopping rule): AUROC of each signal at the stopping round for the correctness of the answer given there (answered stops only). Bold: best per column. Bottom: accuracy of all answered stops and of the half ranked highest by top-token probability or by the uptake direction.

count (synthetic 1–3 hops; for RAGBench, one versus two gold documents). The stored best-layer probe of each source, trained on all groups, is scored per group on test questions without refitting; increment counts are in parentheses.

T The Evidence State Transition Benchmark

The Evidence State Transition Benchmark (ESTB) tests whether the evidence-state readout and its causal profile, established on MuSiQue trajectories in Sections 5 and 6, survive controlled changes to the evidence. It puts all of our sources under one labelling scheme and one evaluation protocol. This appendix describes what the benchmark measures, how each axis was built, and the full results of the retrieval-source and linguistic-form runs, which appear only in summary in Section 7.

T.1 What the benchmark measures

Unit of measurement. As in the rest of the paper, the unit is an *evidence state transition*: an increment $C_{k-1} \rightarrow C_k$ that adds one paragraph (or, for post-sufficiency and full-context increments, a set of paragraphs) to the context, with the paired update $\Delta h = h(C_k) - h(C_{k-1})$ read at the answer-start anchor. ESTB annotates every increment with the *quality state* of the evidence it adds (Table 19) and with the values of the other axes (Table 20): retrieval source, linguistic form, domain, question type, hop count and closed-book status.

Labels. Each increment carries the sufficiency label $y^{\text{suff}} = 1[C_k \text{ sufficient}, C_{k-1} \text{ not}]$, which is defined by construction from the gold hops and does not depend on the model, and a behavioural uptake label for increments that cross sufficiency (answer becomes correct or not). An increment crosses sufficiency exactly when it supplies the last missing hop, whether in the original wording (C3), paraphrased, or from an independently au-

Dataset	grouping	group	sufficiency	uptake	correction
HotpotQA	closed book	incorrect	0.978 (7840)	0.724 (986)	0.839 (5554)
HotpotQA	closed book	correct	0.984 (1689)	0.802 (214)	0.885 (760)
HotpotQA	qtype	bridge	0.978 (7878)	0.680 (982)	0.839 (4963)
HotpotQA	qtype	comparison	0.985 (1651)	0.787 (218)	0.880 (1351)
RAGBench	domain	biomedical	0.929 (1473)	0.776 (291)	0.818 (1331)
RAGBench	domain	finance	0.899 (1081)	0.641 (206)	0.769 (949)
RAGBench	domain	technical	0.881 (1069)	0.675 (196)	0.733 (989)
RAGBench	domain	customer support	0.897 (716)	0.569 (153)	0.658 (587)
RAGBench	domain	general	0.835 (1821)	0.721 (352)	0.702 (1520)
RAGBench	domain	wikipedia	0.978 (729)	0.557 (158)	0.766 (612)
RAGBench	n hops	1	0.836 (2476)	0.698 (444)	0.785 (2211)
RAGBench	n hops	2	0.931 (4413)	0.677 (912)	0.711 (3777)
Synthetic	n hops	1	0.922 (3645)	0.992 (486)	0.875 (2450)
Synthetic	n hops	2	0.970 (6192)	0.928 (1032)	0.926 (4929)
Synthetic	n hops	3	0.970 (11501)	0.916 (2395)	0.931 (8859)

Table 18: Held-out probe AUROC by question group (test-set increment counts in parentheses): closed-book status, question type, domain and hop count.

thored source.

Measurements per cell. A *cell* is the set of increments of one run that share one value on one axis (for example, “E5-retrieved distractors, quality state C1”). For every cell we report:

1. **Within-run readout:** the test AUROC of the run’s own held-out probe (trained on the run’s training questions), for sufficiency, matched sufficiency and uptake. This is the ceiling: is the state linearly decodable at all under this kind of evidence?
2. **Transfer without retraining:** the AUROC of the Qwen MuSiQue two-hop direction (document split, at its best layer), applied unchanged. Because the ESTB runs sample from the same MuSiQue pool as the source run, 818 of the 1,200 questions in the benchmark-gold runs (273 of 402 and 152 of 219 in the fully retrieved runs) are in the source probe’s training split. These questions are *excluded* from every transfer score, so transfer is measured only on questions the source probe never saw.
3. **Quality-state separability:** for each quality state Q , the AUROC separating Q ’s increments (negatives) from the decisive increments of the same run (positives). Near 1, the readout cleanly distinguishes Q from completing the evidence; lower values mean that Q looks like sufficiency to the probe.
4. **Causal profile:** on 150 held-out questions, the answered rate of the insufficient prompt unpatched, with the full sufficient anchor residual patched in, and with only the probe-direction component ($\parallel v$) patched in, all at

every layer from the best one onward; and the answered rate of sufficient prompts after projecting v_{dm} to its insufficient-state value (Section 6).

The benchmark therefore separates three questions: whether the state is *present* (1), whether it is the *same* state as on MuSiQue (2, 3), and whether its causal profile, an inert probe direction alongside a causally sufficient anchor and a v_{dm} abstention gate, carries over (4).

What the benchmark does not measure. ESTB does not score answer quality or retrieval quality for their own sake: retrieval and paraphrasing only vary the evidence that enters a trajectory. It does not ask whether the model *should* answer from partial or contradictory evidence; those states are scored by whether the readout separates them from genuine completion. The sufficiency label is structural (all hops present) rather than semantic, so a passage that happens to state a fact in a way our string-based labelling misses (Appendix T.7) is counted as insufficient.

T.2 Evidence quality states

Table 19 lists the seven quality states of the evidence-quality axis and the increment kinds (Appendix A) that realise them. All states except C0 are increments, i.e. transitions *into* the state from the preceding context.

The contradictory state has two sources. On benchmark and retrieved trajectories, the false value is another question’s sub-answer of the same relation type (false_synthetic); on the synthetic benchmark, it is a generated value of the same type. The tables label the first kind “C5 contradictory (synthetic)” because the false value is

State	Name	Increment kinds (added evidence)	y^{suff}
C0	none	no evidence (start state of every trajectory)	–
C1	irrelevant	distractor_from_none, distractor: a length-matched distractor added to an empty or penultimate context	0
C2	partial	gold_nondecisive: a gold hop that leaves another hop missing; partial: the decisive paragraph rewritten to withhold its value; redundant from a penultimate state (C2+redundant)	0
C3	complete	decisive, loo_decisive: the last missing hop; decisive_paraphrase, decisive_independent: the same hop paraphrased or from an independent source	1
C4	complete + redundant	post_distractor, post_redundant, post_redundant_paraphrase, post_redundant_independent, full_context: further evidence added after sufficiency	0
C5	contradictory	false: the decisive hop with its value replaced (from a penultimate state); post_false: a falsified copy added to a sufficient context	0
C6	answer string, no relation	answer_ctrl: a distractor with “(Not to be confused with {answer}).)” appended	0

Table 19: ESTB evidence-quality states and the increment kinds that realise them. y^{suff} : whether the increment crosses into sufficiency.

substituted rather than found in a real document; naturally occurring contradictions are not separated from near_miss passages by our labelling (Appendix T.7). Partial evidence is defined in two ways: for multi-hop questions, a gold hop that leaves another hop missing (“partial (hop)”); for any question, the decisive paragraph rewritten so that its subject and relation remain but the value is withheld (“partial (masked)”, Appendix T.5).

T.3 Axes and sources

All ESTB runs reported here use Qwen2.5-7B-Instruct (Qwen Team, 2024) and the pipeline of the main text without modification. The four retrieval runs and the variants run draw 1,200 two-hop MuSiQue training questions that the model answers incorrectly closed-book, with no random hop permutations.

T.4 Retrieval-source axis

Corpus and retrievers. Evidence comes from wiki-18, the December 2018 Wikipedia dump of Karpukhin et al. (2020) as distributed with FlashRAG (Jin et al., 2025): 21M passages of 100 words with their article titles. *BM25* (Robertson and Zaragoza, 2009), a lexical retriever, uses the bm25s implementation (Lù, 2024) over title and text, with English stop-words and Snowball stemming. *E5* (Wang et al., 2022), a dense retriever, uses the published FAISS flat inner-product index of e5-base-v2 passage embeddings; queries are encoded with the query: prefix and mean pooling, then ℓ_2 -normalised.

Queries. A single question-level query rarely retrieves both hops of a bridge question. Each question therefore issues the question itself plus one

query per hop, taken from MuSiQue’s decomposition with the earlier hops’ sub-answers substituted for the #k placeholders (three queries per two-hop question). This imitates an iterative or decomposing RAG system that gets the earlier hops right. The per-query hit lists are merged round-robin by rank, de-duplicated, and cut at $k = 30$ passages per question.

Passage labels. Each retrieved passage is labelled against the question’s gold hops by title and string matching (titles normalised by lower-casing and removing a trailing parenthetical):

- **retrieved_gold:** the title matches a gold hop’s page *and* the passage contains that hop’s sub-answer;
- **same_page:** the title matches a gold hop’s page but the sub-answer is absent, i.e. partial evidence found in the wild;
- **near_miss:** not a gold page, but the passage mentions a bridge entity or the final answer, which is the natural analogue of the answer-string control C6;
- **irrelevant:** everything else.

Table 21 gives the label composition. *E5* retrieves at least one gold passage for 85.5% of questions and both hops for 33.5%; *BM25* does so for 68.1% and 18.2%.

Trajectories. Two modes turn retrieval output into trajectories.

- **Retrieved distractors (noise mode):** the benchmark gold paragraphs are kept and every distractor is replaced by a retrieved non-gold passage. The 18 distractors per question

Axis	Levels	Source (runs)
Evidence quality	C0–C6 (Table 19)	every run
Retrieval source	benchmark; BM25; E5 (dense); each as distractors only or as all evidence verbatim; LLM paraphrase; independently authored; templated (5 families)	MuSiQue 2-hop rebuilt over wiki-18 (Appendix T.4)
Linguistic form		MuSiQue variants run (Appendix T.5); synthetic benchmark
Domain	Wikipedia; biomedical; finance; technical; customer support; general web QA	MuSiQue, HotpotQA, 2WikiMultiHopQA; RAGBench’s 12 sub-datasets
Question type	bridge; comparison; single-hop	HotpotQA; RAGBench; synthetic
Hop count	1; 2; 3–4	synthetic, RAGBench; MuSiQue 2-hop; MuSiQue 3–4-hop
Closed-book status	correct; incorrect	HotpotQA and RAGBench (select: all)

Table 20: ESTB axes. The domain, question-type, hop-count and closed-book axes reuse the external runs of Section 7, scored per group in Appendix S; the retrieval-source and linguistic-form axes required new construction and are detailed here.

Retriever	retrieved passages (top 30)				questions with gold for	
	gold	same page	near miss	irrelevant	≥ 1 hop	both hops
E5	0.080	0.139	0.292	0.489	0.855	0.335
BM25	0.053	0.080	0.331	0.535	0.681	0.182
Distractors used in the trajectories (18 per question): same page / near miss / irrelevant						
E5	0.231 / 0.460 / 0.309					
BM25	0.133 / 0.535 / 0.332					

Table 21: Retrieved passages by label (1,200 MuSiQue two-hop questions, multi-query retrieval, $k = 30$).

are taken hardest first: same-page passages, then near misses, then irrelevant passages, each in rank order. The trajectory builder then draws length-matched distractors from this pool exactly as it does from the benchmark pool. Distractors are thus no longer the benchmark’s curated negatives but what a retriever returns for the same question, and nearly a quarter of E5’s distractors come from the gold pages themselves.

- **All evidence retrieved** (*full* mode): the gold paragraph of each hop is also replaced, by the highest-ranked `retrieved_gold` passage for that hop, so every paragraph the model sees is a 100-word wiki-18 passage. Only questions with a retrieved gold passage for both hops can be used: 402 for E5 and 219 for BM25.

Every other part of the construction (chains, controls, falsification, answer-string hatnotes, post-sufficiency and full-context increments) is unchanged, which yields the full set of quality states over retrieved evidence: 20,400 increments in each noise run, 6,809 (E5) and 3,703 (BM25) in the full runs.

T.5 Linguistic-form axis

LLM paraphrase and partial variants. For every gold hop paragraph of the 1,200 questions,

Qwen2.5-7B-Instruct generates two rewrites by greedy decoding (at most 220 tokens):

- *paraphrase*: “Rewrite the following encyclopedia paragraph in completely different wording and sentence structure, keeping every fact exactly the same. You must keep the exact string ‘{sub-answer}’ verbatim.”
- *partial*: “Rewrite the following encyclopedia paragraph so that it still talks about ‘{subject}’ and the same topic, but withholds the specific fact ‘{sub-answer}’: do not mention ‘{sub-answer}’ or any equivalent name or value anywhere; instead say that the detail is not recorded here.”

Generations are validated before use. A paraphrase is kept only if it contains the sub-answer, is between $0.5\times$ and $2\times$ the original’s length, and shares fewer than 60% of its word 4-grams with the original. A partial is kept only if it contains neither the sub-answer, nor the final answer, nor any answer alias, has at least 15 words, and still mentions the hop’s subject. Of 2,400 hop paragraphs, 2,180 (90.8%) yield a valid paraphrase and 1,411 (58.8%) a valid partial.

Independently authored sources. MuSiQue, HotpotQA and 2WikiMultiHopQA all take paragraphs from Wikipedia, but from different dumps,

with different paragraph boundaries and often different revisions. For each MuSiQue gold hop with page title T and sub-answer a , we search the gold and distractor paragraphs of the *other* two datasets for one whose normalised title is T , that contains a , and that shares fewer than 60% of its 4-grams with the MuSiQue paragraph; when several qualify, the least overlapping is used. This finds an independently written statement of the same relation for 228 of the 2,400 hops (144 from HotpotQA, 84 from 2WikiMultiHopQA).

Increments. From the penultimate state, the paraphrased and independent paragraphs give decisive_paraphrase and decisive_independent increments (C3, crossing sufficiency), and the partial paragraph gives a partial increment (C2, which does not). After sufficiency, the paraphrase or independent version of a hop already present gives post_redundant_paraphrase and post_redundant_independent increments (C4). Together with the standard kinds, the variants run has 25,632 increments, including 2,180 paraphrased decisive, 228 independent decisive and 1,411 partial ones.

Synthetic templated form. The synthetic benchmark (Section 7) supplies the fully controlled end of the axis: fictional entities over seven relations (founder, birthplace, capital, spouse, award, employer, headquarters), five template families per relation, one to three hops, and for every gold fact a paraphrase, a partial (value withheld) and a contradictory (different value) variant, with near-miss distractors (same relation, other subject; same subject, other relation) and unrelated filler.

T.6 Results

Retrieval source. With retrieved distractors (Table 22), within-run sufficiency is unchanged (0.946 E5, 0.951 BM25, against 0.948 with benchmark distractors), and the MuSiQue direction transfers at 0.935/0.939 on source-unseen questions. Retrieved distractors are harder than the benchmark’s: the transferred direction separates C1 increments from decisive ones at 0.906 (E5) and 0.917 (BM25), against 0.945 for benchmark distractors, and answer-string controls at 0.962/0.971 against 0.982. The run’s own probe recovers most of this (0.972/0.982 for C1), so the drop is a shift in the distractor distribution rather

than a loss of the state. Uptake falls within run (0.68, against 0.78), but the transferred MuSiQue uptake direction scores 0.817 on the same increments: these runs have 1,200 rather than 2,000 questions, so the within-run probe is trained on fewer decisive increments, which is the likely reason. With all evidence retrieved, sufficiency stays at 0.91–0.92 within run and 0.905–0.910 transferred, and C1 separability is the lowest of any run (0.84 transferred, 0.89–0.92 own), which is expected when every paragraph is a 100-word passage from the same corpus as the gold ones. In every retrieval cell the causal profile of Section 6 reproduces: the full anchor patch raises answering from 0.10–0.30 to 0.68–0.77, the probe-direction component leaves it exactly at the unpatched rate, and projecting out v_{dm} leaves 0.01–0.03 answered.

Linguistic form. Paraphrased and independently authored decisive paragraphs are read as completing the evidence just as the original wording is. Against all non-decisive increments, the transferred MuSiQue direction scores original decisive increments at 0.931 AUROC, paraphrased ones at 0.929 and independently authored ones at 0.918, with mean projections 0.101, 0.098 and 0.096 (non-decisive: -0.082). Paraphrastic and independent redundancy after sufficiency is cleanly separated from completion (0.974 and 0.995 own probe, Table 23). The two hardest quality states are the ones that most resemble completion. Contradictory evidence added from the penultimate state (C5, the decisive paragraph with a swapped value) reaches only 0.76 own and 0.73 transferred, and it has the same “supplies the missing relation” surface form as the decisive paragraph. Masked partial evidence (C2), with subject and relation present but the value withheld, reaches 0.81 own and 0.74 transferred; its mean projection (0.044) lies between the non-decisive and decisive means. Partial evidence in the multi-hop sense (a non-decisive hop) is separated almost perfectly (1.00). Contradictory evidence is the hardest control in every retrieval run as well (0.71–0.75 transferred) and is also the state that projects closest to completion on the synthetic benchmark (Section 7).

Summary. Across the retrieval-source and linguistic-form axes, the sufficiency state remains decodable (≥ 0.91), the MuSiQue direction transfers to unseen questions (≥ 0.90), wording

Evidence source	n_q	within-run probe (AUROC)			MuSiQue dir.		answered rate				
		suff.	matched	uptake	suff.	uptake	unpatched	full	$\parallel v$	proj.-out	v_{dm}
Benchmark gold + benchmark distractors	2000	0.948	0.911	0.775	(source)	(source)	0.13	0.71	0.13	0.01	
Gold + E5-retrieved distractors	1200	0.946	0.909	0.681	0.935	0.817	0.10	0.77	0.10	0.02	
Gold + BM25-retrieved distractors	1200	0.951	0.915	0.679	0.939	0.817	0.10	0.77	0.10	0.01	
All evidence E5-retrieved	402	0.922	0.891	0.839	0.905	0.762	0.19	0.72	0.19	0.03	
All evidence BM25-retrieved	219	0.912	0.876	0.748	0.910	0.820	0.30	0.68	0.30	0.02	
+ LLM paraphrase / partial / independent	1200	0.935	0.885	0.718	0.929	0.796	0.10	0.77	0.10	0.00	

Table 22: Retrieval-source and linguistic-form axes (Qwen2.5-7B, MuSiQue 2-hop). Within-run probes on the document split, except the two fully retrieved runs (question split, too few questions for document components). MuSiQue direction: transfer without retraining, scored only on questions outside the source probe’s training split. Answered rate: the insufficient prompt unpatched, with the full sufficient anchor, and with only the probe-direction component; and the sufficient prompt with v_{dm} projected to its insufficient value.

Quality state vs. decisive increment	n	own probe	MuSiQue dir.
C5 contradictory (synthetic)	764	0.762	0.725
C2 partial (masked)	436	0.809	0.736
C1 irrelevant	1146	0.942	0.941
C4 complete+paraphrase	381	0.974	0.952
C6 answer string, no relation	764	0.975	0.979
C2+redundant	764	0.982	0.978
C4 complete+independent	57	0.995	0.950
C4 complete+irrelevant	382	0.997	0.980
C4 complete+redundant	382	0.999	0.984
C5 complete+contradictory	382	1.000	0.994
C2 partial (hop)	764	1.000	0.996
C4 full context	382	1.000	0.999

Table 23: Quality-state separability (MuSiQue variants run): AUROC separating each state’s increments (negatives) from decisive increments (positives), with the run’s own probe (test questions) and with the MuSiQue direction (source-unseen questions). Lower values mean the state looks more like completing the evidence.

and authorship of the decisive evidence do not matter, and the causal dissociation reproduces in every cell. What changes is the difficulty of the controls: retrieved distractors from the gold pages and value-withheld or value-swapped versions of the decisive paragraph are where a deployed sufficiency detector would err.

T.7 Limitations of the construction

- **Oracle decomposition.** The per-hop retrieval queries substitute gold sub-answers into MuSiQue’s decomposition, so the retriever behaves like an iterative system that solves the earlier hops correctly. Gold coverage is an upper bound on what a realistic multi-hop retriever would achieve.
- **String-based passage labels.** `retrieved_gold` requires the gold page title and the sub-answer string. A passage on another page that states the same relation is labelled `near_miss` and used as a distractor; a gold-page passage that mentions the sub-answer in an unrelated sentence counts

as gold. Both introduce some label noise, in opposite directions.

- **Selection in the fully retrieved runs.** Only questions whose two hops were both retrieved are kept (34% for E5, 18% for BM25), which favours questions with lexically salient hops.
- **Self-generated paraphrases.** Paraphrases and partials are generated by the same model that is probed. Validation removes rewrites that drop the sub-answer or copy the original, but generated text may still be easier for the model to read than human-written text. The independently authored paragraphs avoid this but are few (64 decisive increments on source-unseen questions).
- **One model.** All ESTB runs reported here use Qwen2.5-7B-Instruct.

U Compute and reproducibility

Collection for one 2-hop model ($\approx 35k$ conditions) takes about 22 hours on 64 CPU cores (bf16,

2177 ≈ 300 prompt tokens/s) or about one hour on an
2178 H200 shared with another job; the 3–4-hop run
2179 took 1.3 GPU-hours; all causal experiments and
2180 text baselines together used about 7 GPU-hours.
2181 Probing runs on CPU in one to two hours per
2182 model. Every stage is deterministic given the
2183 seed (question sampling, insertion positions, false-
2184 answer choice, splits) and resumable. Code, con-
2185 figurations and the trajectory construction are re-
2186 leased with the paper.